Billow

A minimum-energy structural model for soft kites

Timoshenko beams, inflatable tubes, wrinkling membranes,

cables and pulleys — one potential energy, one NLP

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Abstract

Billow is a standalone structural library for soft-wing airborne wind energy systems. It assembles cables, frictionless pulleys, geometrically exact Timoshenko beams, inflatable tubes and wrinkling membrane fabric into a single total potential energy, and finds static equilibrium by minimising it with IPOPT. Stationarity of the energy *is* the equilibrium equation, so a converged solve satisfies the nodal force balance to solver tolerance rather than to whatever residual a relaxation scheme happens to leave behind.

This document specifies the formulation, the element library, the validation against benchmarks external to the code, a set of demonstration cases, and a measured comparison against `kite_fem`, the existing FEM path. Headline results: exact-to-10^{-4} agreement with the Euler elastica, second-order convergence to the exact roll-up circle, agreement with published values for the Bathe–Bolourchi 45° bend, reproduction of the ASKITE inflatable-tube fits to 0.06% in bending, and a full-kite structural solve in 5.9 s cold and 0.58 s warm at 1893 degrees of freedom.

Billow currently lives as `awetrim.structural` and imports only NumPy and CasADi. It is intended to become its own repository.

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1 Scope and design intent

1.1 What Billow is

A static structural solver for the kinds of structures soft kites are made of: thin lines that cannot push, ropes that run over sheaves, inflated tubes that bend until they collapse, and fabric that carries tension but wrinkles instead of carrying compression. Given a geometry, material properties and a set of nodal loads, it returns the deformed shape and the internal force distribution.

1.2 What Billow is not

It contains **no aerodynamics**. There are no polars, no airfoils, no lift model. Loads arrive as nodal forces from outside. Everything in this document that involves pressure uses a prescribed uniform differential pressure, not an aerodynamic solution. The same is true of `kite_fem`, which is also purely structural.

It is also not a dynamics code: there are no inertial terms and no time integration. Equilibrium is quasi-static.

1.3 Isolation

Billow depends only on NumPy and CasADi. It does not import the particle-system solver (PSS), the vortex-step method (VSM), the aerostructural coupling loop or any kite YAML schema, and nothing in those modules imports it. This is deliberate and is the first invariant to preserve: it allows the element physics to be validated against closed-form solutions before any of it touches a coupled run. A coupling adapter, when written, belongs on the aerostructural side, not inside Billow.

1.4 Why a new model

The existing minimum-energy solver in the AWETrim tree (`aerostructural/pss/structural_nlp.py`) already solves the current bridle well. It does not extend, for two independent reasons.

Physics. Its element vocabulary is axial springs. Battens and leading-edge tubes need bending and torsion; the canopy needs a membrane that cannot carry compression.

Scale. It builds the objective with a Python loop over elements in CasADi `MX`, so the expression graph grows with the element count. At tens of springs this is free. At tens of thousands of fabric triangles, graph construction — and especially `gradient` and `hessian` over it — becomes the bottleneck long before the numerical solver does.

Billow keeps the formulation (still an NLP, still IPOPT, still minimum potential energy) and replaces the assembly.

2 Formulation

2.1 Total potential energy

The unknown configuration X minimises

$$\Pi(X) \;=\; \sum_{\text{sets}} U_{\text{set}}(X) \;-\; \mathbf{f}_{\text{ext}}\cdot\mathbf{x} \;-\; \mathbf{m}_{\text{ext}}\cdot\boldsymbol{\theta}(\boldsymbol{\psi}) \;+\; \tfrac{1}{2}k_a\|\mathbf{x}-\mathbf{x}_{\text{seed}}\|^2, \tag{1}$$

subject only to bounds on pinned degrees of freedom. The last term is a weak anchor discussed in §2.8.

Because $\partial\Pi/\partial X = 0$ is the equilibrium equation, a converged solve satisfies $\mathbf{f}_{\text{int}} + \mathbf{f}_{\text{ext}} = 0$ at every free node to solver tolerance. There is no separate residual criterion to tune.

2.2 Degrees of freedom

$$X = \big[\ \underbrace{\mathbf{x}_0 \dots \mathbf{x}_{n-1}}_{3 \text{ per node}} \ \big|\ \underbrace{\boldsymbol{\psi}_0 \dots \boldsymbol{\psi}_{r-1}}_{3 \text{ per rotational node}} \ \big] \tag{2}$$

Three translations for *every* node, and three incremental rotations only for the r nodes that some beam element needs a material frame for. Cable, pulley and membrane nodes stay at three degrees of freedom. A canopy of 10 000 fabric nodes therefore costs 30 000 unknowns, not 60 000, regardless of what beams are attached elsewhere. For contrast, `kite_fem` allocates six degrees of freedom per node everywhere and masks the unused ones.

2.3 Finite rotations: the Cayley parameterisation

Rotations use the **Rodrigues (Cayley) vector** $\boldsymbol{\psi} = 2\tan(\phi/2)\,\mathbf{n}$ rather than the exponential map's rotation vector $\phi\,\mathbf{n}$. With $S = \operatorname{skew}(\boldsymbol{\psi})$,

$$R(\boldsymbol{\psi}) = I + \frac{4}{4 + \boldsymbol{\psi}\cdot\boldsymbol{\psi}}\left(S + \tfrac{1}{2}S^2\right), \tag{3}$$

$$\boldsymbol{\psi}(R) = \frac{2\,\operatorname{axial}(R - R^{\mathsf{T}})}{1 + \operatorname{tr} R}, \tag{4}$$

$$\boldsymbol{\psi}_{1/2} = \frac{\boldsymbol{\psi}}{1 + \sqrt{1 + \tfrac{1}{4}\boldsymbol{\psi}\cdot\boldsymbol{\psi}}}, \qquad R(\boldsymbol{\psi}_{1/2})^2 = R(\boldsymbol{\psi}). \tag{5}$$

The motivation is entirely numerical. Every map above is a *rational* function of its argument. There is no $\sin\phi/\phi$ removable singularity, so no `if_else` branch is needed at $\phi = 0$, so the Hessian that IPOPT differentiates is continuous everywhere in $|\phi| < \pi$. The half-angle identity (5) is worth noting: the naive form $\tan(\phi/4) = (\sqrt{1+t^2} - 1)/t$ has a removable 0/0 at $t = 0$; rationalising it removes the singularity exactly.

2.4 Incremental rotations and frame updates

$\boldsymbol{\psi}_j$ is measured against a stored reference frame, so the configuration is $R_j = R(\boldsymbol{\psi}_j)\,R_{\text{ref},j}$ and every solve starts from $\boldsymbol{\psi} = 0$. After each solve the increment is absorbed, $R_{\text{ref}} \leftarrow R(\boldsymbol{\psi}^\star)R_{\text{ref}}$, and re-projected onto $SO(3)$ so repeated absorption cannot drift.

The Cayley chart is singular at $\phi = \pi$. Incremental rotations mean a node can turn arbitrarily far *across a load ramp* while each individual solve stays well inside the chart. This is why applied moments require load stepping (§2.5); nodal forces do not.

2.5 Applied moments are not conjugate to $\boldsymbol{\psi}$

A moment does work against the *exponential-map* rotation vector, not against the Rodrigues vector the degrees of freedom carry. The external work term therefore converts,

$$\boldsymbol{\theta}(\boldsymbol{\psi}) = \boldsymbol{\psi}\,\frac{2\arctan(\|\boldsymbol{\psi}\|/2)}{\|\boldsymbol{\psi}\|}, \tag{6}$$

which tends to $\boldsymbol{\psi}$ as $\|\boldsymbol{\psi}\| \to 0$.

This is not cosmetic. Using $-\mathbf{m}\cdot\boldsymbol{\psi}$ directly is correct only to first order, and the roll-up benchmark (§4.2) exposed it as a *load-path dependence*: the same static problem gave $8.1\times10^{-3}L$ with four load steps and $2.1\times10^{-1}L$ with eight. Statics must be path independent. With the conversion in place the answer is identical to five digits across 4, 8 and 16 steps, and this is guarded by a regression test.

Two limitations remain, both fundamental rather than implementation artefacts:

- Applied moments need load stepping, because the chart is singular at $\phi = \pi$. Four steps suffice for a full 2π roll-up.
- A constant ("dead") moment in three dimensions is **genuinely non-conservative** and therefore has no potential at all. Energy minimisation can represent only the fixed-axis case.

Nodal *forces* — which is what an aerodynamic coupling applies — carry none of these caveats and are exact in a single solve.

2.6 Assembly: one mapped kernel per element type

This is the single design decision that makes the model scale. Each element kernel is compiled once to a small CasADi `SX` function over *one* element and evaluated across the whole set with `Function.map` on a gathered index matrix:

$$Q = \text{reshape}\big(X[\,\text{idx}\,],\ n_{\text{dof/el}},\ n_{\text{el}}\big), \qquad U_{\text{set}} = \textstyle\sum_e \mathcal{U}(Q_{:,e}, F_{:,e}, P_{:,e}).$$

The objective graph therefore holds one node per element *type*, not per element. Graph construction, gradient and Hessian generation cost $O(\text{number of element types})$, not $O(\text{number of elements})$.

The Hessian retains the sparsity of the element connectivity graph, so IPOPT's sparse linear algebra sees a mesh-stencil matrix and scales accordingly. The measured consequence is Table 1.

Table 1: Flat clamped canopy, uniform out-of-plane load. Iteration count is essentially mesh independent; cost sits in the sparse linear algebra. Source: `run_demo_cases.py --case scaling`.

Triangles	DOF	Build (s)	Solve (s)	IPOPT iterations
128	243	0.07	0.05	6
1800	2883	0.67	1.25	9
5832	9075	3.0	6.1	9

2.7 Everything numeric is a parameter

Reference frames, every element parameter column (rest lengths, stiffnesses, section properties), the external loads and the anchor are NLP *parameters*, not baked constants. One solver build therefore serves an entire sweep: tape actuation, a stiffness ramp or a load sweep change parameter values, never topology.

2.8 The anchor term

The final term of (1) is a weak spring to the seed positions, present only to pin a node that no taut element reaches and whose equilibrium position would otherwise be indeterminate. It biases the reported force balance by at most $k_a\|\mathbf{x} - \mathbf{x}_{\text{seed}}\|$; the default $k_a = 10^{-6}\,\text{N/m}$ puts that far below any coupling tolerance.

2.9 Convergence criterion

`converged` is the *physical* verdict: the solve is accepted when the free nodes are in force balance to

$$\max\big(\varepsilon_{\text{abs}},\ \varepsilon_{\text{rel}} \max_i \|\mathbf{f}_{\text{ext},i}\|\big), \qquad \varepsilon_{\text{abs}} = 10^{-6}\,\text{N},\ \ \varepsilon_{\text{rel}} = 10^{-6},$$

whether or not IPOPT liked its own termination test. The relative term matters: an absolute newton means very different things on a 1 N bridle line and a 600 N benchmark beam. IPOPT's own verdict is preserved alongside as `ipopt_success` and `status` — on stiff, nearly quadratic structures it routinely reports `Search_Direction_Becomes_Too_Small` at an answer already exact to machine precision.

3 Element library

Every kernel returns the strain energy of one element. Kernels are pure: no loops over elements, no global indexing, no closing over per-element data. Everything element-specific arrives through the parameter row.

3.1 Cable

A linear axial spring with stretch $s = \ell - \ell_0$:

$$U = \tfrac{1}{2} k \langle s \rangle^2, \qquad \langle s \rangle = \max(0, s) \text{ when tension-only.} \tag{7}$$

The tension-only cut is C^1 but not C^2: the stiffness jumps across the taut/slack boundary. An optional `slack_smoothing` width rounds the corner to recover a continuous Hessian at the cost of a small spurious compressive stiffness.

3.2 Pulley

A rope running over a frictionless sheave at the middle node, with a single shared stretch across both arms:

$$U = \tfrac{1}{2} k \left\langle \|\mathbf{x}_2 - \mathbf{x}_1\| + \|\mathbf{x}_3 - \mathbf{x}_2\| - \ell_0 \right\rangle^2. \tag{8}$$

One shared stretch is exactly the statement that rope tension is equal either side of the sheave, and it lets the pulley node find its own position along the rope.

Convention warning. `rest_length` here is the **whole rope**, both arms. This is not universal: the PSS geometry reader splits ℓ_0 across the two arms, while `kite_fem` stores the total on each arm. Any geometry adapter must state which convention it is reading.

3.3 Geometrically exact Timoshenko beam

Two-node Simo–Reissner beam with one-point (mid-element) integration. Each node carries a position and an orthonormal material frame $R = [\mathbf{d}_1\, \mathbf{d}_2\, \mathbf{d}_3]$: twelve degrees of freedom per element.

With $\boldsymbol{\psi}$ the relative rotation between the two node frames from (4), and R_m the exact half-way frame from (5):

$$\boldsymbol{\Gamma} = R_m^{\mathsf{T}} (\mathbf{x}_2 - \mathbf{x}_1)/L_0, \tag{9}$$

$$\boldsymbol{\Omega} = R_m^{\mathsf{T}} \boldsymbol{\psi}/L_0, \tag{10}$$

$$U = \frac{L_0}{2} \Big[\Delta\boldsymbol{\Gamma}^{\mathsf{T}} \operatorname{diag}(EA, GA_2, GA_3)\, \Delta\boldsymbol{\Gamma} + \Delta\boldsymbol{\Omega}^{\mathsf{T}} \operatorname{diag}(GJ, EI_2, EI_3)\, \Delta\boldsymbol{\Omega} \Big]. \tag{11}$$

$\boldsymbol{\Gamma}$ is a *deformation* measure rather than a strain: its first component is the axial stretch, which is unity in the reference configuration, and the other two are the transverse shears. $\boldsymbol{\Omega}$ holds the torsion and the two bending curvatures. What drives the energy is the change against the reference, $\Delta(\cdot) = (\cdot) - (\cdot)_0$, so an initially curved or pre-twisted member — a leading-edge tube, a strut — is stress-free as built.

Three properties make this element usable inside an NLP objective:

Objective. Under a superposed rigid rotation Q, both $\boldsymbol{\psi}$ and $\mathbf{x}_2 - \mathbf{x}_1$ map to $Q(\cdot)$ while $R_m \to QR_m$, so $\boldsymbol{\Gamma}$ and $\boldsymbol{\Omega}$ are invariant and rigid-body motion costs exactly zero energy. A slack member that swings freely is not spuriously stiffened.

Shear-locking free. The single midpoint sample is the standard reduced integration cure. Kite battens and leading-edge tubes are slender, so this is not optional; §4.5 verifies it at slenderness 800.

Branch-free. Using the Cayley vector as the curvature measure makes the element strain $\boldsymbol{\Omega}_{\text{exact}} + O(\phi^3)$ per element — the same order as the one-point quadrature error, and vanishing under mesh refinement. This is a deliberate trade: the exponential map's $\sin\phi/\phi$ would need a branch at $\phi = 0$.

3.4 Inflatable tube

An inflated leading-edge tube is not linear-elastic. Its bending stiffness falls away as the compressed side wrinkles, until the section carries a limiting moment and then collapses. `kite_fem` captures this with empirical fits (coefficients $C_1 \dots C_{19}$, from the ASKITE work) applied as *secant* stiffnesses: EI and GJ are recomputed each iteration from the current deflection and twist.

That is consistent inside a Newton residual, but it is **not a potential**. Substituting a state-dependent $EI(\kappa)$ into $\frac{1}{2}EI\kappa^2$ silently drops the $\mathrm{d}EI/\mathrm{d}\kappa$ terms, so the gradient would no longer be the internal moment and a minimum-energy solve would converge to the wrong equilibrium. The fits must be *integrated*, not substituted.

From tip-load fit to moment–curvature law. The bending fit is published as tip load against normalised tip deflection $v = \delta/L$ of a **one-metre** cantilever: $P(v) = D\big(1 - e^{-(N/D)v}\big)$. For a linear cantilever $\delta = PL^3/3EI$ and $\kappa_{\text{root}} = PL/EI$, so $\kappa = 3v/L$, and the root moment is $M = PL$. At the calibration length $L = 1\,\text{m}$ this converts the fit into an intrinsic constitutive law:

$$M(\kappa) = M_{\max}\left(1 - e^{-EI_0\kappa/M_{\max}}\right), \qquad EI_0 = N/3, \quad M_{\max} = D, \tag{12}$$

$$W_b(\kappa) = M_{\max}\left[\,|\kappa| - k_0\left(1 - e^{-|\kappa|/k_0}\right)\right], \qquad k_0 = M_{\max}/EI_0. \tag{13}$$

W_b is quadratic ($\frac{1}{2}EI_0\kappa^2$) near zero, so it is smooth there despite the $|\kappa|$, and saturates at $M_{\max}$.

Recasting this way also removes a length inconsistency: `kite_fem` infers $EI = P/(3v)$, which equals the true EI only for a one-metre element, because v is already normalised by element length. A moment–curvature law is length-independent by construction.

Torsion is already intrinsic (twist per unit length) and integrates directly:

$$T(\omega) = c_1 \arctan(c_2\omega), \qquad W_t(\omega) = c_1\left[\omega\arctan(c_2\omega) - \frac{\ln\big(1 + c_2^2\omega^2\big)}{2c_2}\right], \tag{14}$$

with $GJ_0 = c_1c_2$ and a limiting torque $c_1\pi/2$.

Biaxial bending uses the resultant curvature $\sqrt{\Omega_2^2 + \Omega_3^2}$, the correct isotropic extension for an axisymmetric tube. Axial and shear stiffness are *not* covered by the fits and stay linear; supply them from the tube geometry.

Table 2: Derived tube properties from the ASKITE fits. Higher pressure stiffens and strengthens; the thinner tube collapses *before* its moment saturates ($\kappa_{\text{collapse}} < k_0$).

d (m)	p (bar)	EI_0 (N m^2)	$M_{\max}$ (N m)	k_0 (m^{-1})	κ_{collapse} (m^{-1})	GJ_0 (N m^2)	$T_{\max}$ (N m)
0.16	0.3	288.2	58.10	0.2016	0.2751	91.3	124.9
0.16	0.5	362.0	98.78	0.2728	0.2973	86.5	174.7
0.12	0.3	126.8	34.16	0.2694	0.2217	33.2	117.1

Collapse is reported, not enforced. A genuinely dropping post-collapse moment would make the energy *decrease* with curvature — an unbounded mechanism that minimisation would simply run away from. The law therefore stops at saturation, and `inflatable_beam_state` returns a `utilisation` ratio $\kappa/\kappa_{\text{collapse}}$ and a `collapsed` flag, so leaving the calibrated range is visible rather than silent.

3.5 Wrinkling membrane

Constant-strain triangle with a Saint-Venant–Kirchhoff plane-stress law, in the finite-strain (Green–Lagrange) measure so a canopy panel can rotate and billow arbitrarily without spurious stiffening. Translation-only: a membrane carries no bending.

The reference triangle is flattened once into its own in-plane basis and stored as the inverse edge matrix D_0^{-1}, so the kernel only ever sees a 2×2 map:

$$F = [\,\mathbf{a}\;\mathbf{b}\,]\,D_0^{-1}, \quad C = F^{\mathsf{T}}F, \quad E = \tfrac{1}{2}(C - I), \tag{15}$$

$$W_{\text{taut}} = \frac{E_{\text{mod}}}{2(1-\nu^2)}\left[(\operatorname{tr} E)^2 - 2(1-\nu)\left(E_{11}E_{22} - E_{12}^2\right)\right]. \tag{16}$$

Wrinkling. A membrane cannot carry compression. Left alone, the SVK law lets the canopy mesh buckle into whatever compressive mode the discretisation offers, and the result depends on mesh density rather than on physics. The relaxed (tension-field) energy in principal Green strains $e_1 \geq e_2$ is the classical Pipkin form:

$$W_{\text{relaxed}} = \begin{cases} W_{\text{taut}}, & e_2 + \nu e_1 \geq 0 \quad \text{(taut)}, \\ \tfrac{1}{2}E_{\text{mod}}\, e_1^2, & e_1 > 0 \text{ otherwise} \quad \text{(wrinkled)}, \\ 0, & e_1 \leq 0 \quad \text{(slack)}. \end{cases} \tag{17}$$

This is exactly what a minimum-energy formulation wants: the relaxed functional is the quasiconvex envelope, so the minimiser lands on the wrinkled state directly instead of chasing the near-singular tangent a residual-form Newton solve has to fight through. **Solving fabric by energy minimisation is not a workaround — it is the better-posed formulation.**

Slack fabric needs a residual stiffness. A fully slack region stores *exactly* zero energy, so its Hessian block is exactly zero and the Newton step is undefined; IPOPT reports `Error_In_Step_Computation` on a panel clamped into a frame smaller than itself. Blending a small fraction of the unrelaxed law back in,

$$W = (1-r)\,W_{\text{relaxed}} + r\,W_{\text{taut}}, \qquad r = 10^{-4}, \tag{18}$$

leaves the taut region untouched (there the two coincide) while giving slack fabric a definite tangent. It is also not a fiction: real fabric has some compressive and bending resistance. Do not set $r = 0$ on a canopy that can go slack.

3.6 Uniform pressure

A ram-air cell, a single skin pinned along its edge, or a tube modelled with its own fabric is loaded by a uniform gauge pressure p inside the same minimisation. The work of a uniform pressure is p times the swept volume, so its potential is

$$U = -p\,V, \qquad V = \sum_t \tfrac{1}{6}\,\mathbf{x}_1 \cdot (\mathbf{x}_2 \times \mathbf{x}_3), \tag{19}$$

the enclosed volume written as a sum of signed cone volumes from the origin, one translation-only three-node element per triangle, mapped like any other kernel. For a node whose fan of triangles is closed, $\partial V/\partial \mathbf{x}_i$ is the vector area of the ring polygon around it, $\frac{1}{3}\sum_t A_t \mathbf{n}_t$: exactly the force a lumped $pA\mathbf{n}/3$ scheme applies, at every interior node. Positive p inflates an outward-wound surface; the builder checks that the surface is a closed, consistently wound manifold enclosing a positive volume, and names the first offending edges when it is not.

Open surfaces. The cone volumes depend on where the origin is; their sum over a closed surface does not, and neither does the gradient at any node whose fan is closed. An open surface (a disk pinned at its rim, a cylinder between rigid rings) is therefore also a genuine potential provided every rim node is pinned: only the rim carries origin-dependent terms, and a pinned node's gradient never enters the equilibrium.

One scalar per set. The exactness at interior nodes holds only for a pressure that is uniform over the node's fan. With a per-triangle p_t the sum $-\sum_t p_t V_t$ is no longer translation-invariant (measured: a change of 1.3×10^{-13} N under a translation for uniform p, 20.9 N for a per-triangle field), so its gradient is not a pressure force at all. The pressure therefore stays a parameter column, so a ramp runs on one solver build, but it is one scalar per element set: the builder refuses an array and the force diagnostic refuses a column that is not constant over each connected piece of the surface. Several cells at different pressures are several sets, one closed surface each; a rib shared by two cells appears in both with opposite windings and carries the net $p_{\text{left}} - p_{\text{right}}$, and at equal pressure an internal rib belongs in no pressure set at all.

What this kernel is not. A flow-dependent pressure distribution, the aerodynamic Δp over a single skin, is non-uniform and has no potential. That load goes through `lumped_pressure_forces`, a NumPy helper that lumps $p_t A_t \mathbf{n}_t/3$ at the nodes, applied by the adapter as a dead load lagged by one coupling step, as drag is in the quasi-steady tether example. For uniform p the lagged iteration and the potential share their fixed point, which `test_pressure.py` checks. A fixed-mass gas law, $-p_0 V_0 \ln(V/V_0)$, is a function of the global volume rather than of one element and is not provided; an open ram inlet is a free boundary and not conservative.

Three consequences for a solve. The triple product is odd under a reflection, so the mirrored half of a symmetric surface must be wound in reversed order (the "even in each strain" argument of the symmetry constraints does not apply; the orientation check catches a half copied without the flip). The solver's acceptance tolerance scales with the external forces only, so a pressure-only solve is judged at the absolute `force_tolerance` against nodal pressure forces of order pA/n_{nodes} and the tolerance must be scaled to match. And `StructuralSolution.strain_energy` is the total element energy including $-pV$, so an inflated balloon reports a large negative number. Finally, the SVK membrane and its relaxed form both grow like λ^4 in a uniform stretch while pV grows like λ^3, so the energy of an inflated balloon is bounded below and a one-shot solve is well posed; the limit-point instability of a neo-Hookean balloon is real physics this constitutive law cannot show.

4 Validation

All references in this section come from *outside* the code: an exact boundary-value solution, an exact circle, published tabulated values, or the independent fits themselves. Nothing here can drift with the implementation.

Where `kite_fem` appears it is driven with matched linear section properties and its inflatable constitutive update disabled, so the two share one material model. Section properties are deliberately slender: the classical references are inextensible and shear-rigid, so axial and shear flexibility are kept below $\sim 0.1\%$ and cannot be mistaken for discretisation error.

4.1 Euler elastica

The *elastica* is the classical large-deflection equilibrium shape of a thin, inextensible, shear-rigid rod — the problem Jacob Bernoulli posed and Euler solved in 1744, whose exact solution is expressible in elliptic integrals. It is a **reference solution, not a beam formulation**: it generalises nothing about the element used here, and no beam element is involved in producing

it. Its value as a benchmark is that it stays exact at arbitrarily large rotation, which is precisely where a geometrically exact element has to be tested and where a linearised one fails.

The case is a tip-loaded cantilever at load parameter $\alpha = PL^2/EI$ up to 10, which takes the rod through most of a quarter turn. The reference comes from solving the elastica equation $\theta'' = -\alpha \cos\theta$, $\theta(0) = 0$, $\theta'(1) = 0$ as a boundary-value problem to 10^{-10} with continuation in α.

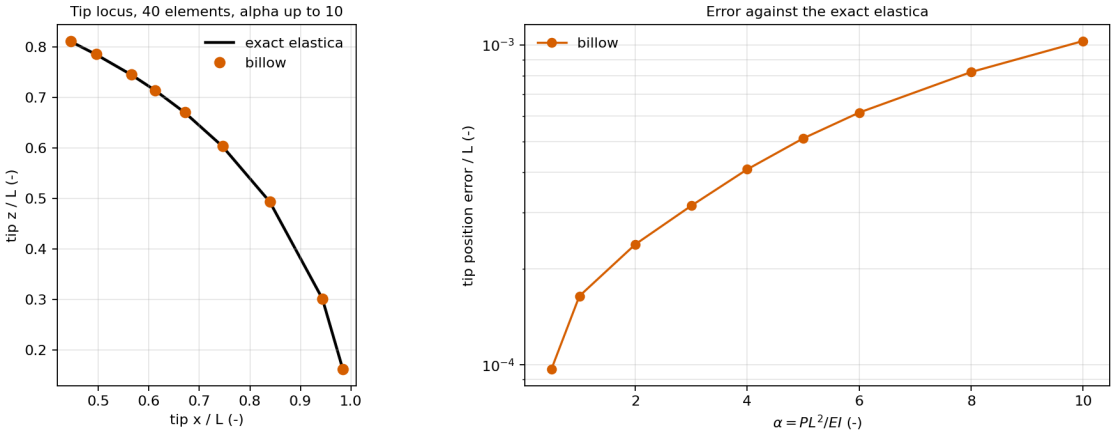


Figure 1: Tip locus and error against the exact elastica, 40 elements. Billow tracks the exact curve across the whole range; `kite_fem` departs progressively and its tip deflection saturates beyond $\alpha \approx 2$.

Table 3: Tip position error normalised by length, 40 elements.

α	0.5	2	6	10
Billow	9.0×10^{-5}	2.4×10^{-4}	6.5×10^{-4}	1.1×10^{-3}
`kite_fem`	1.6×10^{-2}	1.8×10^{-1}	4.2×10^{-1}	4.9×10^{-1}

Mesh refinement separates discretisation from formulation. At $\alpha = 3$, refining from 10 to 160 elements:

Elements	10	20	40	80	160
Billow	3.8×10^{-3}	7.0×10^{-4}	3.1×10^{-4}	4.6×10^{-4}	5.0×10^{-4}
`kite_fem`	2.908×10^{-1}	2.914×10^{-1}	2.916×10^{-1}	2.917×10^{-1}	2.917×10^{-1}

The `kite_fem` error is **mesh independent**, which places it in the formulation rather than the discretisation. Its beam matches linear theory to 0.02% below about one degree of tip rotation, so the property matching used here is correct; the error is purely rotation-driven. Envelope: below 1% up to roughly 10°, about 15% at $\alpha = 1$, saturating beyond $\alpha = 2$. It reports `converged=True` at those answers.

4.2 Roll-up under an end moment

A cantilever under $M = 2\pi EI/L$ must roll into an exact closed circle, so the reference requires no lookup at all.

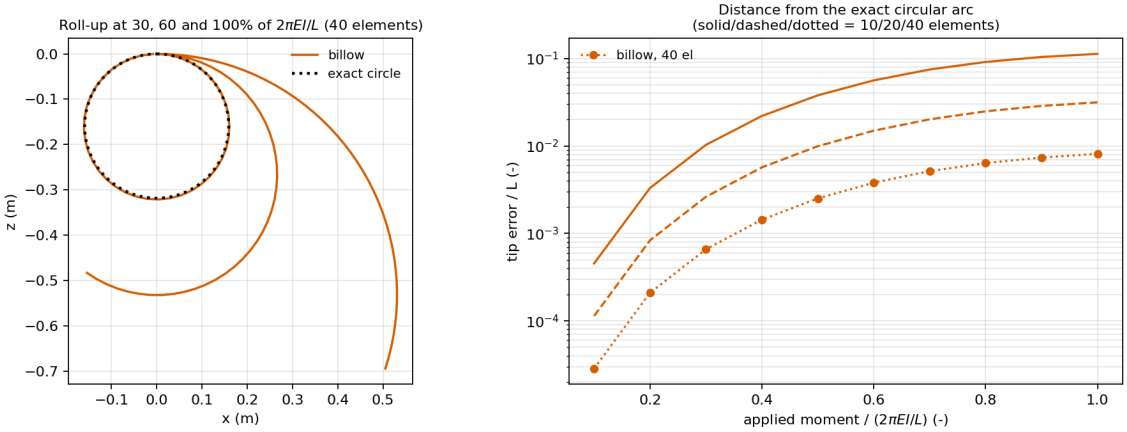


Figure 2: Roll-up at 30, 60 and 100% of $2\pi EI/L$, and the distance from the exact circular arc. Billow converges at second order; `kite_fem` moves away from the exact answer under refinement.

Table 4: Tip distance from the exact arc, normalised by length.

Elements	10	20	40
Billow (4 load steps)	1.13×10^{-1}	3.16×10^{-2}	8.14×10^{-3}
`kite_fem` (single shot)	5.90×10^{-2}	1.71×10^{-1}	2.32×10^{-1}

Billow's error ratios are 3.6 and 3.9, i.e. second order. Extending to 80 elements gives 2.05×10^{-3}, confirming the trend. This benchmark is what exposed the moment-conjugacy defect of §2.5.

4.3 Bathe–Bolourchi 45° bend

The standard three-dimensional finite-rotation benchmark: a 45° circular arc of radius 100, square section, out-of-plane tip load $F = 600$. Published tip *displacements* differ by about 1% between authors, so the reference is a band.

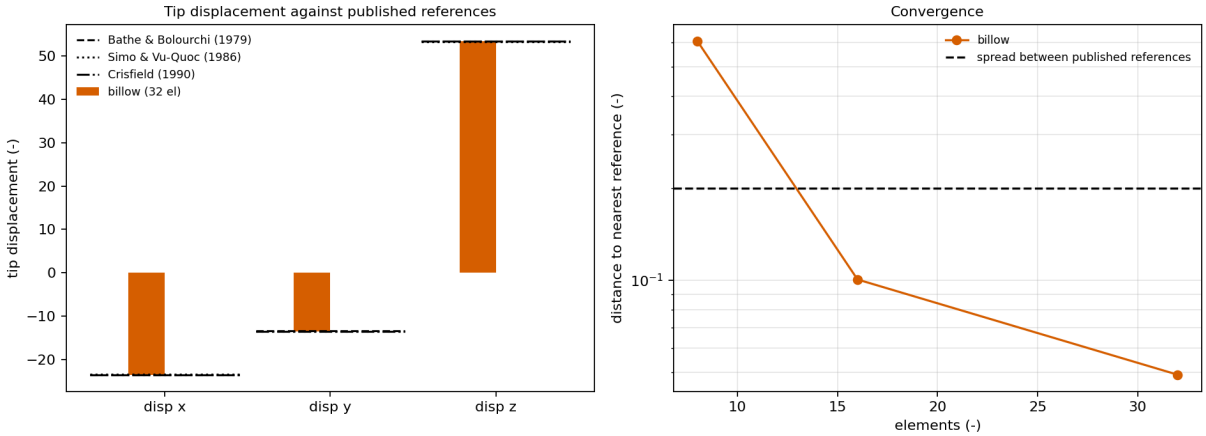


Figure 3: Tip displacement against published references, and convergence.

Table 5: Tip displacement, 32 elements. The arc here starts at the origin with its tangent along $+x$ and sweeps towards $+y$, which swaps the two in-plane components relative to the usual tabulation.

	u_x	u_y	u_z	$\|\mathbf{u}\|$
Simo & Vu-Quoc (1986)	−23.48	−13.50	53.37	59.85
Bathe & Bolourchi (1979)	−23.5	−13.4	53.4	59.85
Crisfield (1990)	−23.5	−13.6	53.4	59.88
Billow	**−23.52**	**−13.60**	**53.44**	**59.95**
`kite_fem`	−37.74	−22.14	28.17	52.04

Billow sits within the spread between published references (0.17% on the magnitude). `kite_fem` is 47% low out of plane.

4.4 Inflatable tube law

The port of §3.4 is verified with **pure end moments**, where the exact solution is a constant-curvature arc at whatever curvature the fit prescribes. That check is exact at any deflection — unlike the tip-load form the fit was published in, whose own inversion assumes linear cantilever theory and therefore cannot serve as a verification instrument.

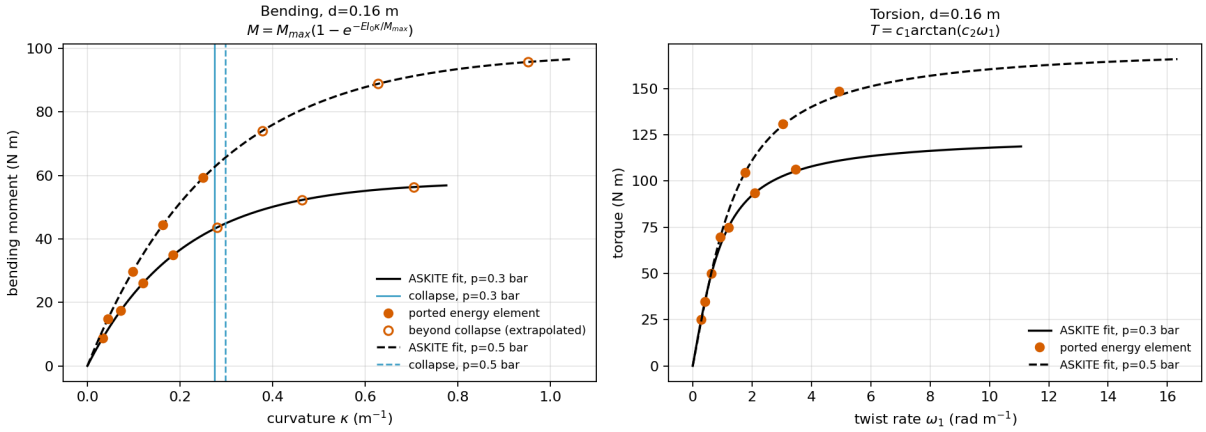


Figure 4: ASKITE fits (lines) with Billow end-moment solves on top (markers). Hollow markers lie beyond the collapse curvature, where the fit is extrapolation: the solve is fine, the calibration is not. This figure completes `kite_fem/examples/FEM_beam_verification.py`, which plots the fitted curves but leaves the simulation loop as comments.

Worst errors: 0.031% (bending, 0.3 bar), 0.057% (bending, 0.5 bar), 0.749% and 1.491% (torsion).

4.5 Beam convergence and shear locking

Tip deflection of a slender cantilever must approach $PL^3/3EI + PL/\kappa GA$ at second order:

Elements	2	4	10	20	40	80
Error vs Timoshenko	−6.25%	−1.56%	−0.250%	−0.063%	−0.016%	−0.004%

The error falls by a factor of four per refinement. Locking is checked separately by holding the load and lengthening the beam through slenderness ratios of 50, 200 and 800: the tip deflection tracks beam theory to 0.2% throughout, where a locking element would stiffen towards zero.

4.6 Membrane

Checked against closed form rather than against another code:

- **Rigid-body motion** costs zero energy under arbitrary rotation and translation — what the finite-strain measure buys over a small-strain one.
- **Equibiaxial stretch** matches the analytic SVK plane-stress energy to 10^{-9} relative.
- **Wrinkled branch** is exactly uniaxial: stretching along x while squashing along y gives $\frac{1}{2}EtAe_1^2$, and squashing further changes nothing.
- **Slack panel** stores exactly zero, where the unrelaxed law would store large energy.
- **Flat-plate inflation** converges under mesh refinement from a configuration with zero out-of-plane stiffness — the singular start that a residual-form Newton method has nothing to invert at.

4.7 Inflated membranes

The uniform-pressure potential, checked against closed forms and a boundary-value solution (`validation/run_pressure_benchmarks.py`).

Exact polyhedra. For a vertex-transitive equilateral polyhedron (tetrahedron, octahedron, icosahedron) of SVK fabric, uniform scaling is the symmetric critical point of the discrete energy, and the stretch is the root of

$$A_0\, tE\, \frac{\lambda^2 - 1}{1 - \nu} = 3\, p\, V_0\, \lambda, \tag{20}$$

with A_0 and V_0 the polyhedron's own area and volume. Free balloons, no node pinned, held only by the anchor: the solved radii about the centroid match λR_0 to 4×10^{-14}.

Balloon. The continuum SVK sphere inflates by the λ that makes $\Pi(\lambda) = \pi R_0^2 tE(\lambda^2-1)^2/(1-\nu) - \frac{4}{3}\pi p R_0^3 \lambda^3$ stationary, $\lambda^2 - a\lambda - 1 = 0$ with $a = pR_0(1-\nu)/(Et)$. An icosphere converges onto it at second order: relative errors 1.27×10^{-3}, 3.34×10^{-4}, 8.27×10^{-5} and 2.05×10^{-5} at subdivision levels 1–4 (80 to 5120 triangles), ratios 3.81, 4.03, 4.05 per refinement. A pressure sweep of ten values runs on one solver build, well under a second, through the parameter column (Fig. 5).

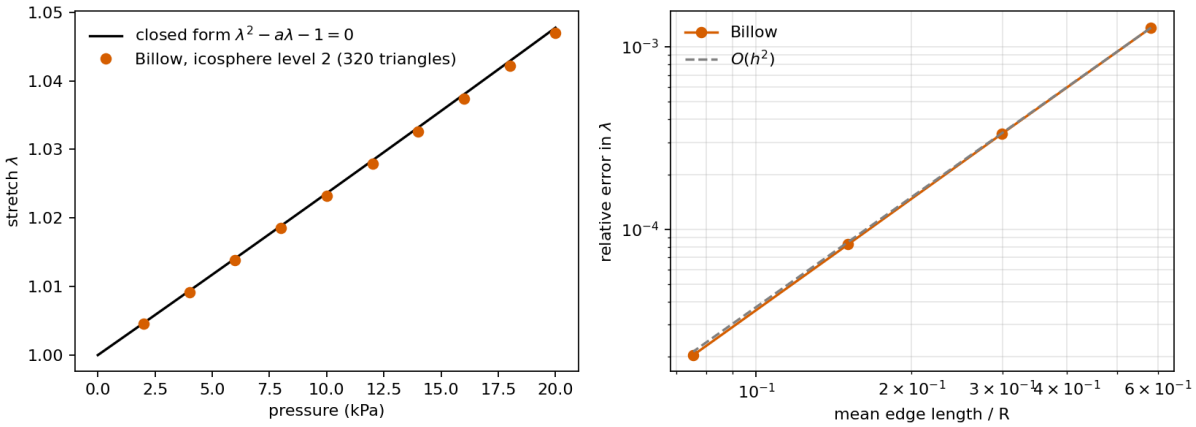


Figure 5: SVK balloon under uniform pressure. Left: stretch against pressure on one solver build, against the closed form. Right: relative error against mean edge length at icosphere levels 1–4, with an $O(h^2)$ guide.

Hencky problem. The Föppl–von Kármán clamped circular membrane under uniform pressure, solved here as a boundary-value problem with `solve_bvp` at $\nu = 0.3$: $w_0/a = 0.65344\,(pa/Eh)^{1/3}$, with the radial stress resultant N_r falling from 0.4311 at the centre to 0.3330 at the rim in units of $Eh\,(pa/Eh)^{2/3}$. The textbook 0.662 is the uncorrected series value. At $pa/Eh = 10^{-4}$ the CST disk converges onto the BVP from above, 2.8×10^{-3}, 9.2×10^{-4} and 5.4×10^{-4} relative at 8, 16 and 24 rings, and the N_r profile taken from the membrane's own Green strains lies on the BVP curve (0.4312 at the centre at 24 rings). Beyond $pa/Eh \approx 10^{-3}$ the coefficient drifts upward with load: finite strain leaving the small-strain theory, not discretisation (Fig. 6).

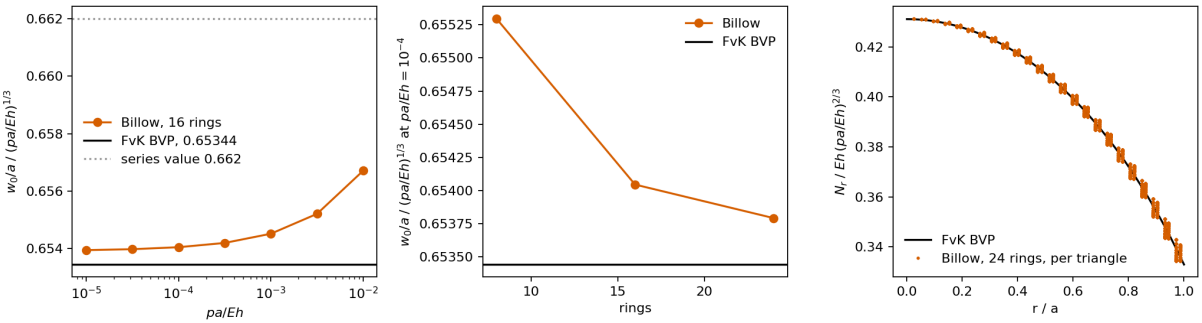


Figure 6: Hencky problem. Left: the deflection coefficient against load at 16 rings, with the BVP value and the uncorrected series value. Centre: refinement at $pa/Eh = 10^{-4}$. Right: radial stress resultant per triangle against the BVP profile.

Cylinder between rigid rings. There is no clean closed form at finite length (an $O(R)$ membrane boundary layer at each ring), so this is a figure rather than an assertion: away from the ends the hoop stretch is the plane-strain form $\lambda_\theta^2 - 1 = 2pR(1-\nu^2)/(Et)$, matched to 1.7×10^{-5} at mid-length for $L/R = 4$ and 8, with the boundary layer about $0.26R$ wide (Fig. 7).

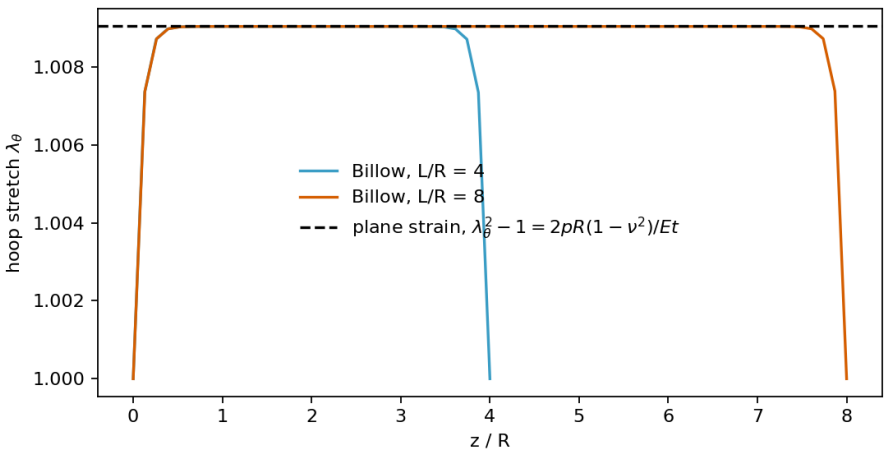


Figure 7: Open cylinder between pinned rigid end rings under internal pressure: hoop stretch along the axis for $L/R = 4$ and 8 against the plane-strain closed form.

4.8 Test suite

158 tests, all passing, organised one file per concern:

`test_rotations.py`	Cayley round trips, half-angle identity
`test_cable.py`	Hooke parity with PSS, slack cut, pulley tension equalisation
`test_beam.py`	objectivity, $O(h^2)$ convergence, no locking, torsion vs GJ
`test_membrane.py`	analytic SVK, three wrinkling branches, inflation, scale
`test_pressure.py`	$-pV$ gradient vs lumped force, exact polyhedra, balloon $O(h^2)$, Hencky BVP, lagged lo
`test_inflatable.py`	energy integrates the fitted moment; end-moment solves
`test_benchmarks.py`	roll-up vs exact circle, load-step independence, bend45
`test_energy.py`	mapped assembly vs a naive per-element sum; DOF layout

The most important of these is in `test_energy.py`: the total energy computed through the gather-and-map machinery is compared against the same energy assembled one element at a time in plain Python, on a deliberately irregular model containing every element type with a beam running over a scattered, out-of-order subset of the node numbering. That machinery is easy to get subtly wrong — a column-major reshape, a rotational slot off by one — and wrong in a way that still converges to a plausible-looking shape.

5 Validation against a measured kite

The benchmarks of Section 4 are exact solutions of idealised problems. This section is the only place where Billow meets a real structure with measured deformation: the TU Delft V3 suspended from a bridle bar in a hangar, at two inflation pressures and five loadings, with its deformed shape recorded by stereoscopic photogrammetry.

Measurements, the reference model and `kite_fem`'s published results are vendored under `data/LEI-V3-KITE/hanging_validation/` from `awegroup/kite_fem` (MIT, © 2025 Patrick Roeleveld), `main` at `630c8c0`. The three-dimensional marker clouds come from Haanen's repository.[1] Thesis references are to P. I. H. Roeleveld, *Fast finite element modelling of bridled leading-edge inflatable kites*, MSc thesis, TU Delft, February 2026. Everything here is regenerated by the `run_hanging_*` scripts in `scripts/structural/`.

Three results carry beyond this experiment. The distributed reference model is *ill posed* — its anchor node belongs to no element — so any future use of the dataset must supply a suspension. `kite_fem`'s beams converge to about 0.23× its own constitutive law, so it is not a clean reference solution. And the dataset is governed almost entirely by inflatable-tube bending, not by the canopy it was chosen to test.

5.1 The experiment

The kite hangs from its bridle and the deformed shape is recorded stereoscopically, with markers on the inflatable structure. Twelve lengths are extracted per load case: nine trailing-edge segments `La`–`Li` between consecutive struts, the tip-to-tip span `b`, and two tip-to-centre-leading-edge diagonals `LcsTL`, `LcsTR`.

The deformation is large. Under self-weight alone the span falls from 8.305 m as built to a measured 4.867 m — a 41% closure — and every trailing-edge segment shortens by 26–43%. The tip-load cases run the same deformation backwards: spreading the tips with 2 kg or 5 kg reopens the span to 7.8–8.3 m, close to the as-built shape.

Suspension. The kite hangs from a **bar**, not a point. Roeleveld's thesis (Appendix B, Table B.1) marks `A_main`, `A_I`, `M`, `PL` and `SL` as *N/A* for the hanging test and adds a `bar` of 1650 mm; Section 6.3 explains that the bridle was adapted to fit the hangar height. The same table records the one documented shortening, `B_r,5` from 12 168 to 9425 mm.

[1] `https://github.com/pimjhaanen/photogrammetry_thesis`

Provenance caveat. The upstream repository describes the measurements only as "experimental validation reference data" and never uses the word photogrammetry; the thesis identifies the method as stereoscopic camera measurement. The published rows are also perfectly symmetric left-to-right to four decimals, so they have been mirrored or port/starboard-averaged.

5.2 Rebuilding the model

Billow is rebuilt from `hanging_test_initial.npz`, the fully assembled reference model, rather than from the YAML — there is no committed YAML→npz builder, so the `.npz` *is* the input both codes solve. Nodes, connectivity, rest lengths, stiffnesses and tube diameters therefore transfer with no re-interpretation:

shared, element for element	276 nodes, 46 bridle cables, 14 pulleys, 98 inflatable tube beams
changed	canopy: 721 tension-only springs → 392 wrinkling membrane triangles

Two checks make this trustworthy rather than assumed. The load vector reproduces `kite_fem`'s own `loading()` to **exactly zero** for all five load combinations, and the length extractor reproduces its published `model_results.csv` from its saved states to **machine zero**.

Canopy grid. The reference canopy is a structured 29×8 grid stored only as a flat spring list. It is recovered geometrically — the reference builder projects every interior section node onto the straight line joining its leading- and trailing-edge node, so each chordwise section is exactly collinear — and then *verified*: all 29 sections must come out with 8 nodes, and all 721 canopy springs must be grid edges with no spring joining two grid nodes being anything else. Both hold exactly.

5.3 Defects found in the reference model

5.3.1 The anchor is connected to nothing

Node 0 is flagged fixed but appears in no spring, no pulley and no beam; seven further bridle nodes are orphaned with it (all of exactly zero mass). The kite is therefore a free body under 126 N of unbalanced weight, and its potential is unbounded below. Billow reports that faithfully: the structure translates hundreds of kilometres, and neither load stepping nor constitutive continuation helps, because none of them supplies the missing constraint.

`kite_fem` does not diverge only because its `I_stiffness` regularisation acts as a ground spring on every node. Its own published solutions still drift 0.29–0.34 m bodily upward, with node 0 sitting fixed and unconnected throughout.

5.3.2 Bridle lengths disagree with the thesis

Twenty-eight of the thirty-six declared bridle lines match Table B.1 to the millimetre, which is what makes the remaining eight look like transcription errors rather than naming mismatches: `BRI` and `BRII` are an exact swap; `abcd2` and `abcd3` carry a duplicated leading digit; `ab3` equals `ab2`; `a5` and `br5` differ; and `brmain ext` is present in the table but unwired.

The eighth is not a transcription error but a contradiction inside the thesis itself. Table B.1 marks `A_I` as N/A for the hanging test, while Figure B.1 *draws* `A_I` running to the bar in the adapted geometry. The two readings are structurally different — cutting the line removes a front load path — so both were run:

Table 6: Three readings of the same source. Cutting `A_I` is the most accurate reading on the cases the model can solve and the worst overall, because removing that load path is what lets the four centre-load cases fall into a folded local minimum. No reading is adopted here: the ambiguity is a finding about the source data, not something to settle by taking whichever scores best.

bridle reading	all ten (%)	six solvable (%)	four centre-load (%)	folded cases
bridle as stored	15.1	11.0	21.3	0
Table B.1, A_I cut	17.3	8.9	30.0	4
Table B.1, A_I kept	15.7	11.5	21.9	0
`kite_fem`	12.4	7.8	19.3	0

5.3.3 The reference beams converge far softer than their own law

Billow's integrated tube law agrees with `kite_fem`'s secant EI to 1–3% across the whole curvature range, departing only below its 0.03 normalised-deflection floor. The two codes therefore implement the same constitutive law. They do not converge to the same stiffness:

load case	median EI (N m^2)	vs. own law	min EI	beams flagged collapsed
1	24.5	0.29×	2.7	78
4	69.4	0.81×	−42.4	49
6	43.1	0.50×	−164	76
9	84.1	0.98×	−459	40

Three of the four cases contain *negative* bending stiffness, which the secant form $EI = PL^3/\big(3(v - PL/kGA)\big)$ produces as soon as the deflection estimate falls below the shear term. `kite_fem` therefore lands closer to the measurement than Billow because its beams are several times softer than the law it implements, not because its physics is closer.

5.4 Results

Table 7: All ten load cases. Errors are the mean over the twelve measured lengths of $|L_{\text{model}} - L_{\text{meas}}|/L_{\text{meas}}$; the reference `shapecorrelation` metric is not used, because as a dot-product ratio dominated by the two quantities of order 4–8 m it reads 0.980–0.9997 even where the span is 48% wrong. $\|r\|$ is the force residual on the worst free node.

	load case			span (m)			mean rel. error (%)		Billow	
LC	p (bar)	tip (kg)	centre (kg)	meas.	Billow	`kite_fem`	Billow	`kite_fem`	t (s)	$\|r\|$ (N)
1	0.15	0	0	4.867	7.276	5.382	25.9	13.7	7	2.2e-07
2†	0.15	0	9.7	5.042	7.554	7.462	27.6	25.0	15	9.1e-08
3†	0.15	0	25.2	5.113	7.841	7.576	21.8	18.9	19	3.8e-07
4	0.15	2	0	7.843	8.597	8.467	6.5	6.7	16	1.1e-07
5	0.15	5	0	8.271	8.916	8.962	5.9	5.4	18	4.1e-07
6	0.25	0	0	5.453	7.468	6.363	19.5	9.3	6	4.0e-09
7†	0.25	0	9.7	5.575	7.690	7.691	19.7	18.9	15	3.7e-09
8†	0.25	0	25.2	5.697	7.965	7.638	18.4	14.6	17	6.4e-09
9	0.25	2	0	7.818	8.593	8.461	5.2	5.6	15	1.1e-07
10	0.25	5	0	8.224	8.908	8.955	6.2	5.9	16	4.0e-07
						mean, all ten	15.7	12.4	14	
						mean, excluding centre-load cases	11.5	7.8	13	

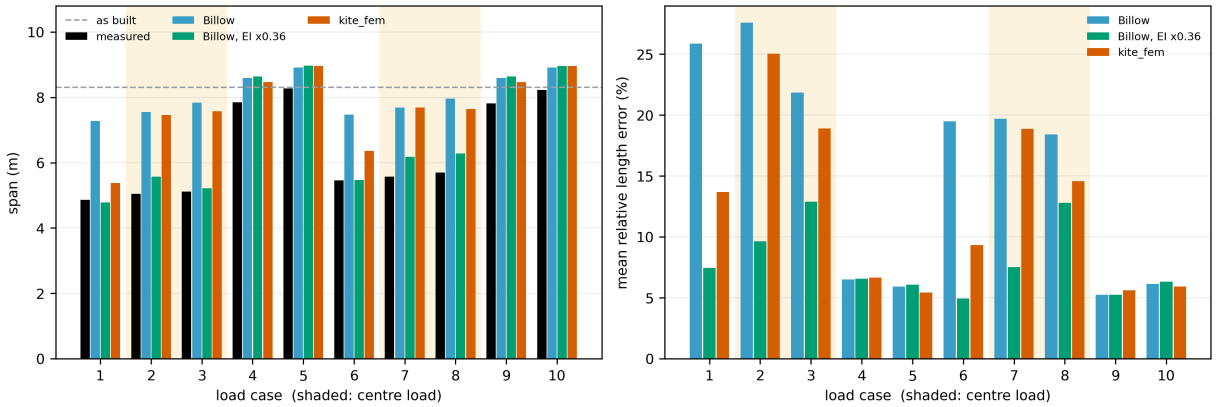


Figure 8: Span and mean relative error across the ten cases, for Billow with its own tube law, Billow with the whole moment–curvature curve scaled by 0.36 (Section 5.9), and `kite_fem`. Both codes overstate the span wherever they solve the physical branch; the gap is widest where gravity alone sets the shape and narrowest where an imposed tip load dominates. The shaded centre-load cases are reproduced by neither code, and are where Billow's folded branch shows as a near-zero span.

Table 8: Per-quantity error, averaged over the ten cases. The two codes fail differently, which the signed-bias columns separate from the magnitude columns: Billow's bias is one-signed — it overstates essentially every length, i.e. its wing stays too open — whereas `kite_fem` scatters, understating the outboard segments and the tip-to-centre diagonals while overstating the inboard ones. A single stiffness factor could plausibly account for the first and could not account for the second.

quantity	meas. range (m)	|rel. error| (%)		signed bias (%)	
		Billow	`kite_fem`	Billow	`kite_fem`
`La`	0.848–0.903	2.4	2.4	+1.9	-0.6
`Lb`	0.979–1.280	16.5	10.6	+16.5	+8.3
`Lc`	0.903–1.276	18.8	14.3	+18.8	+9.7
`Ld`	0.749–1.259	24.5	20.8	+24.5	+20.8
`Le`	0.920–1.217	13.0	16.2	+13.0	+16.2
`Lf`	0.749–1.259	24.1	20.8	+24.1	+20.8
`Lg`	0.903–1.276	18.4	14.3	+18.4	+9.7
`Lh`	0.979–1.280	16.9	10.6	+16.9	+8.3
`Li`	0.848–0.903	2.4	2.4	+1.9	-0.6
`b`	4.867–8.271	30.3	22.9	+30.3	+22.9
`LcsTL`	4.029–4.888	10.3	6.8	+10.3	+5.1
`LcsTR`	4.029–4.888	10.4	6.8	+10.4	+5.1

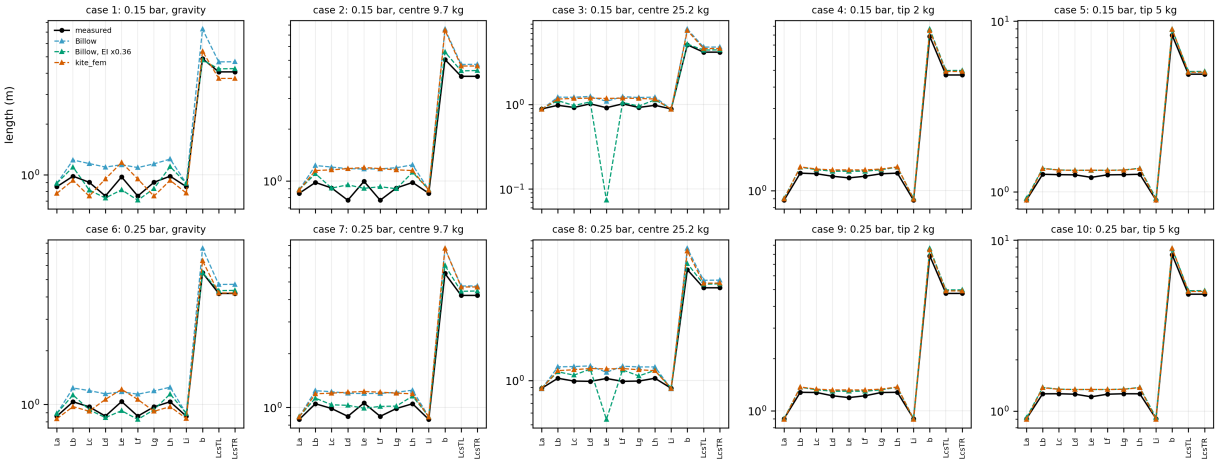
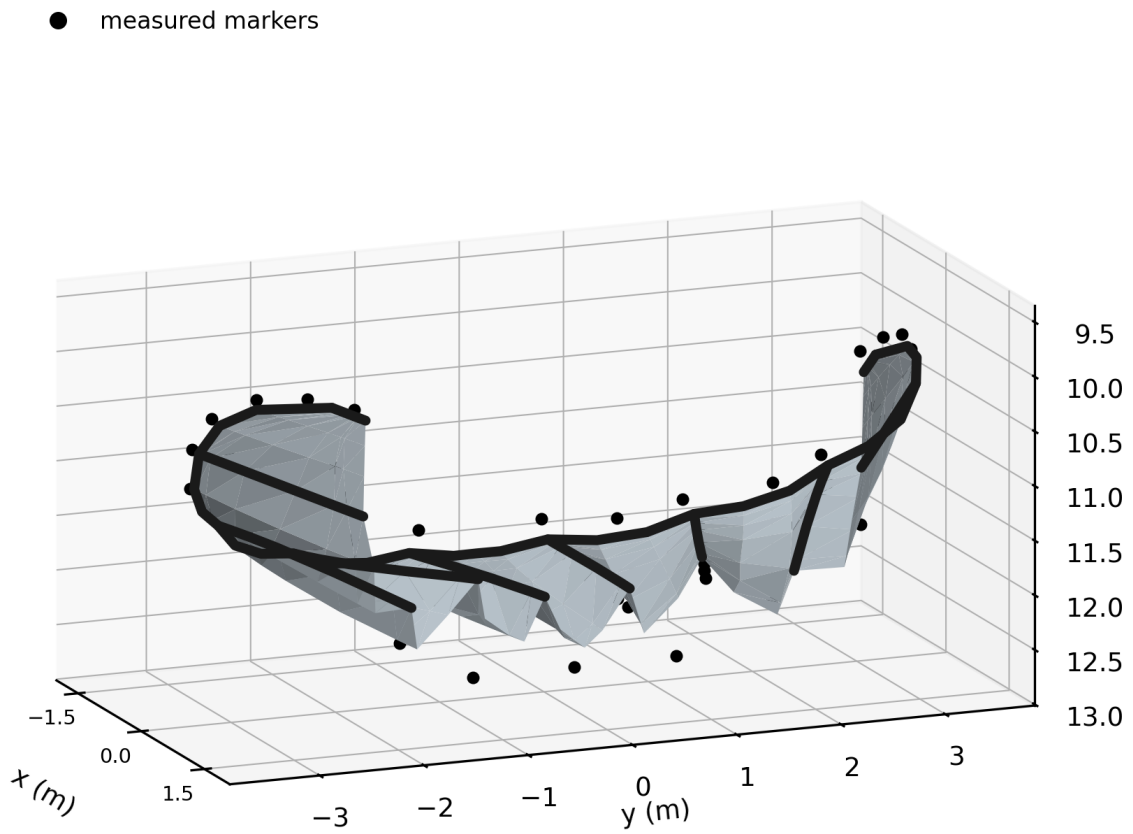


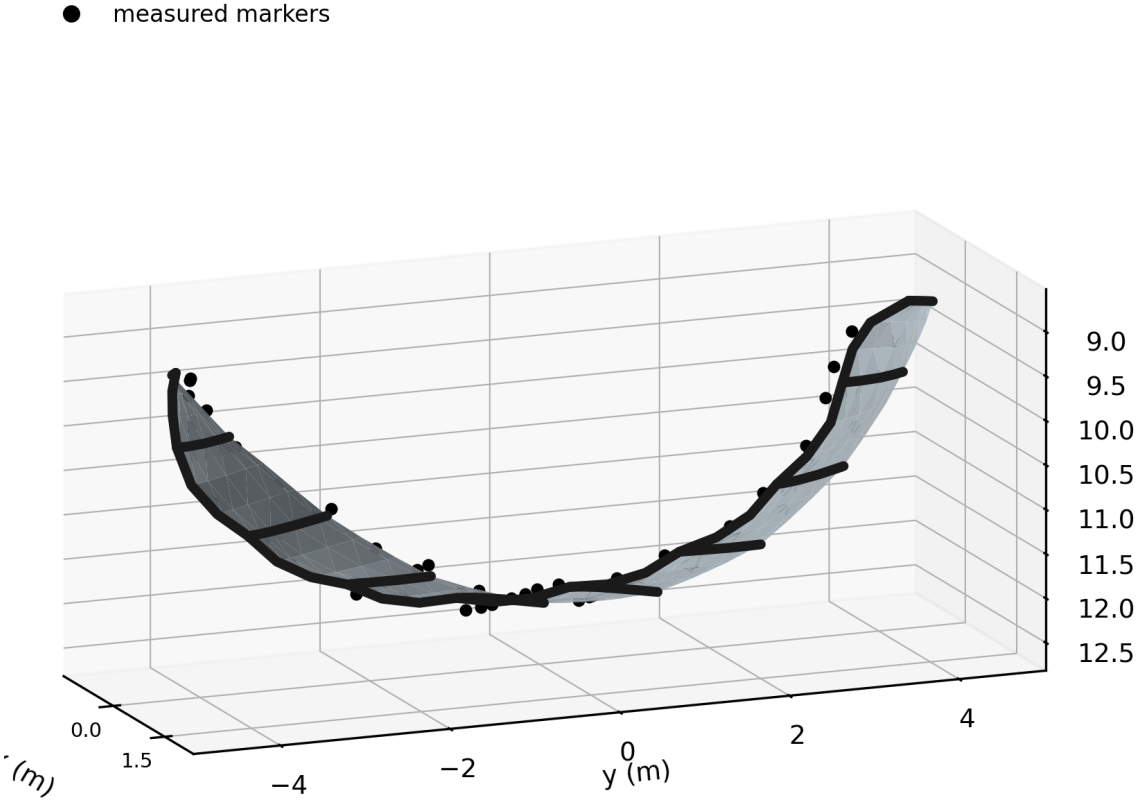
Figure 9: The twelve measured quantities per load case, model over measurement.

5.4.1 The solved kite

case 1: 0.15 bar, gravity | solved shape, measured markers overlaid



case 5: 0.15 bar, tip 5 kg | solved shape, measured markers overlaid



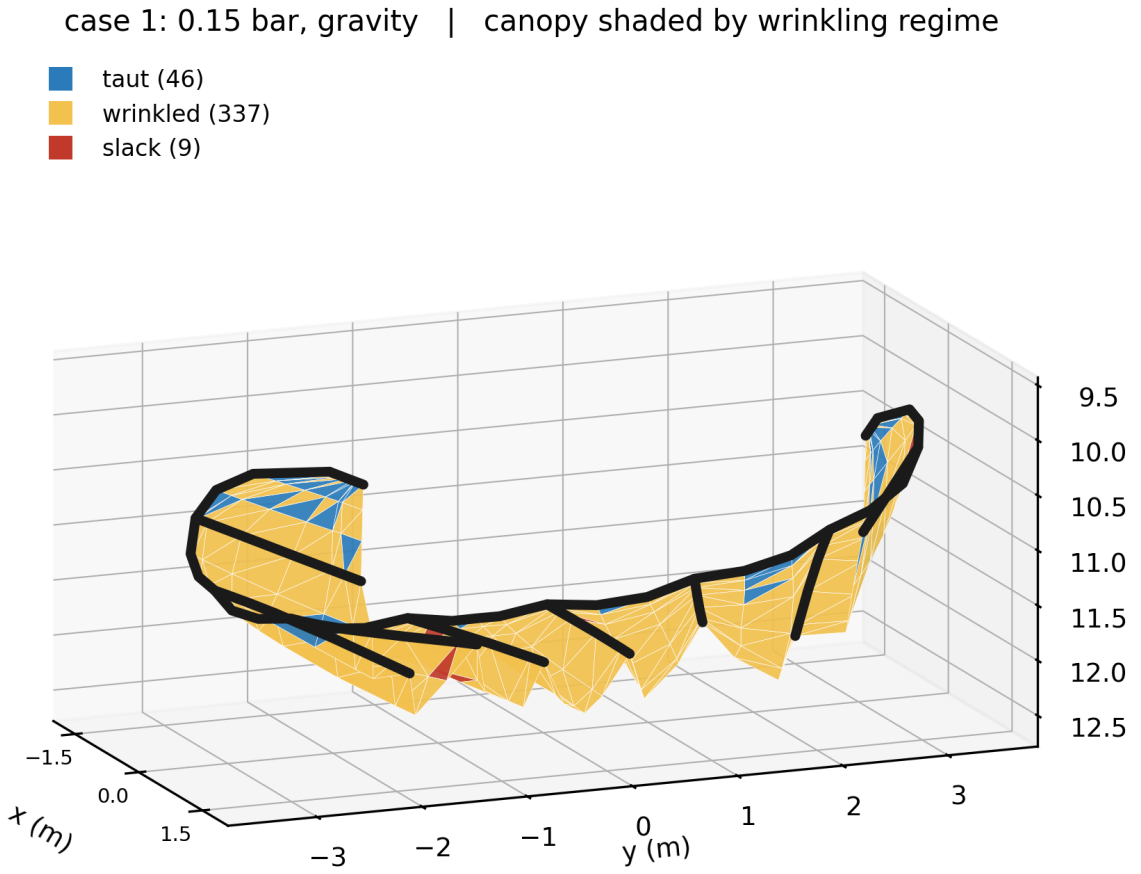


Figure 11: The same self-weight case with each canopy triangle coloured by its tension-field state. This is a prediction the membrane formulation makes and a tension-only spring net cannot: under self-weight almost the whole canopy is wrinkled — one principal stress relaxed to zero — with taut material only where the bridle pulls through, and a handful of fully slack triangles at the tip.

5.4.2 Shape, case by case

Each figure shows the inflatable structure — leading edge and the ten struts — in front, top and side projection, against Haanen's photogrammetry markers: filled on the tubes, hollow on the canopy. Every model is placed by *its own* rigid best fit to the markers (Section 5.9) and then drawn in one common frame, so no model is shown in a frame chosen to suit another. The as-built shape is the dashed null model.

The lower panel says *where* each model is wrong: the distance from every marker to each model's surface, against arc length along the measured leading edge. Arc length rather than span, because on the gravity cases the measured tips curl inward past where any model reaches, and spanwise position would fold back on itself there. Strut markers stand in a column where their strut meets the leading edge, canopy markers at the nearest leading-edge marker. The markers stop short of the tips because the outermost bays are not instrumented.

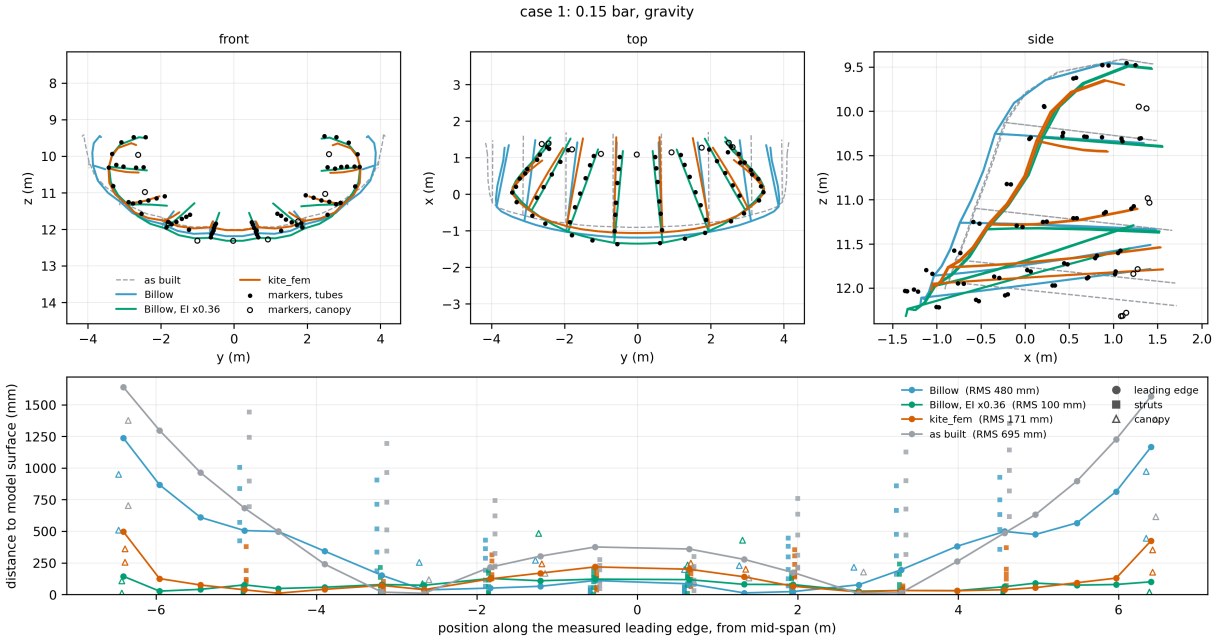


Figure 12: Load case 1.

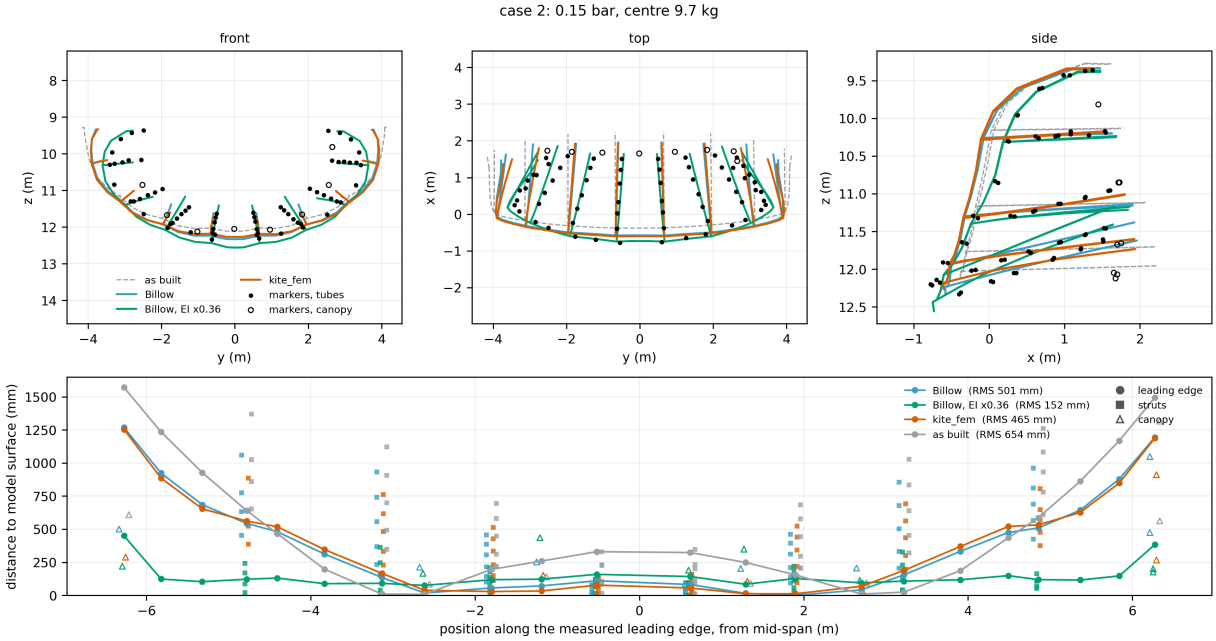


Figure 13: Load case 2.

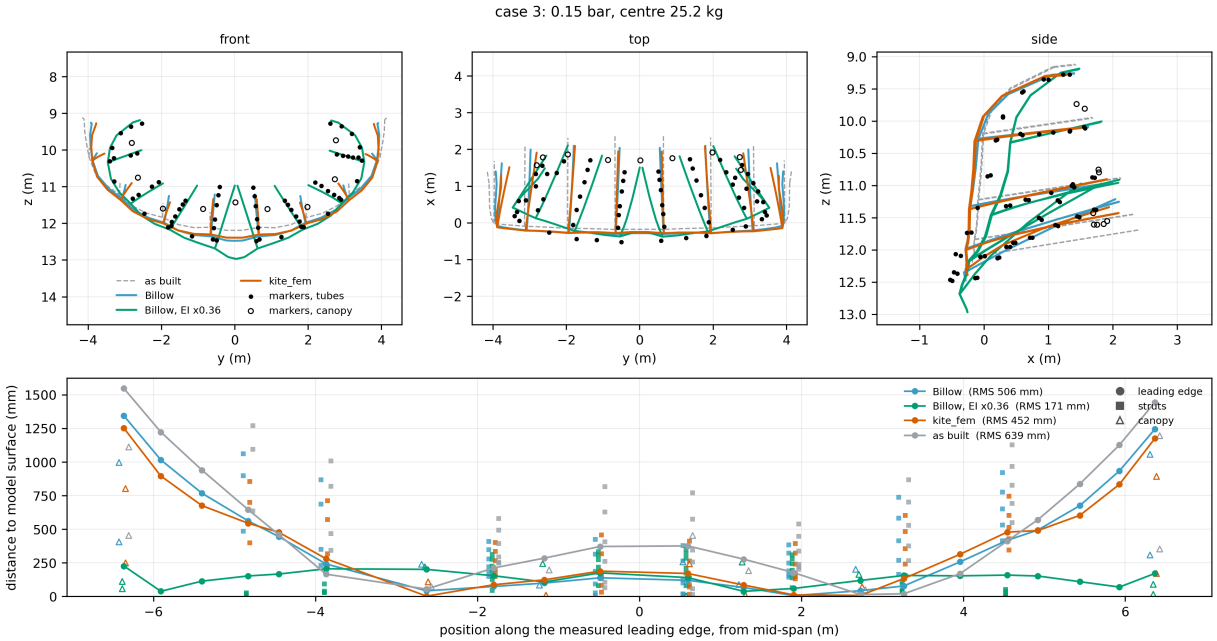


Figure 14: Load case 3.

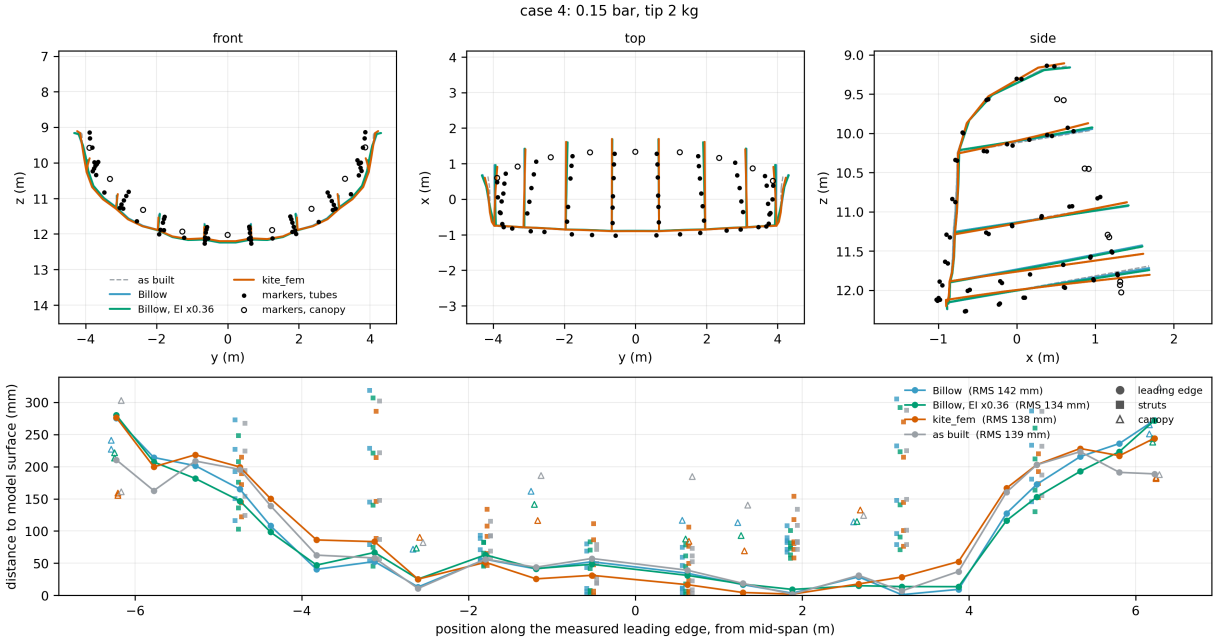


Figure 15: Load case 4.

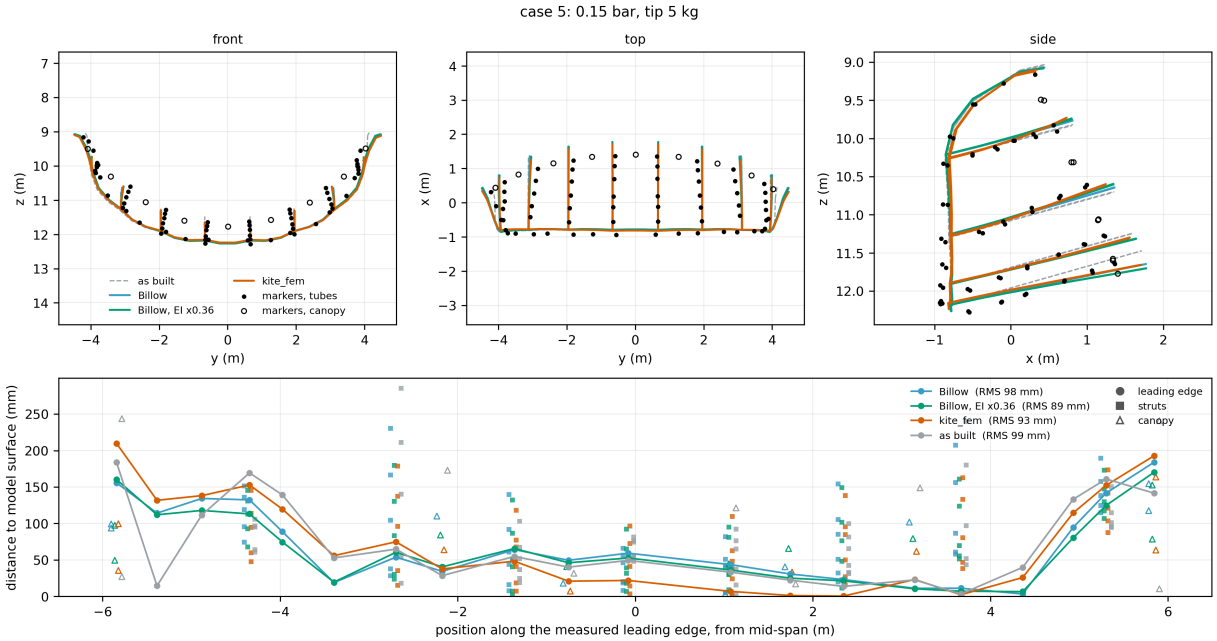


Figure 16: Load case 5.

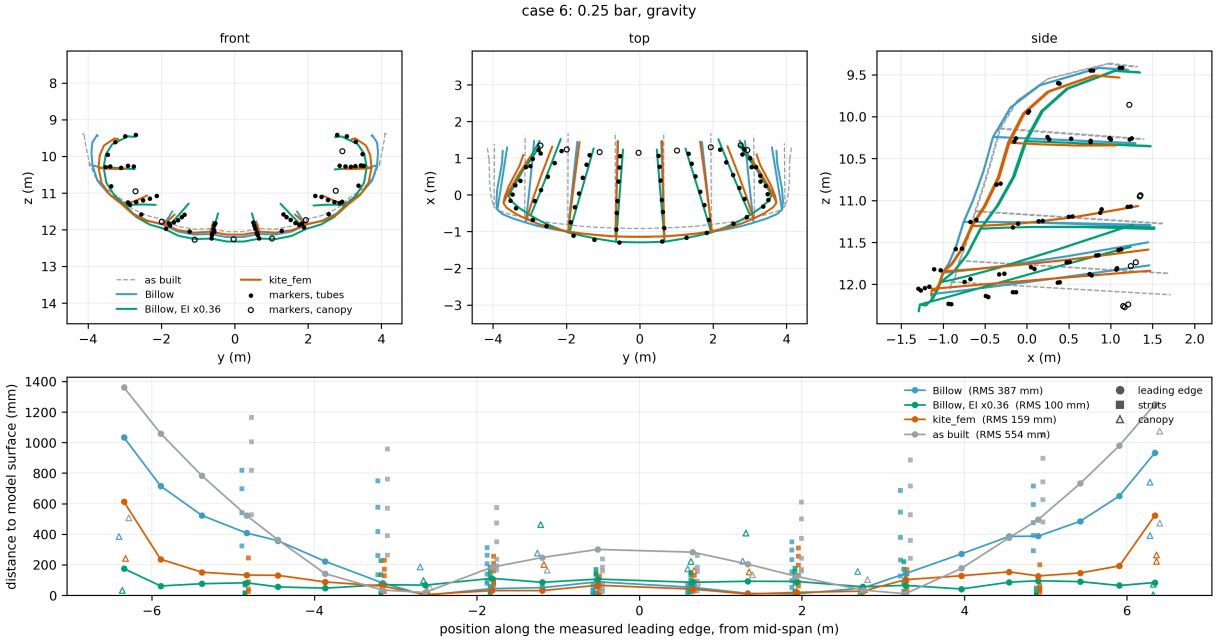


Figure 17: Load case 6.

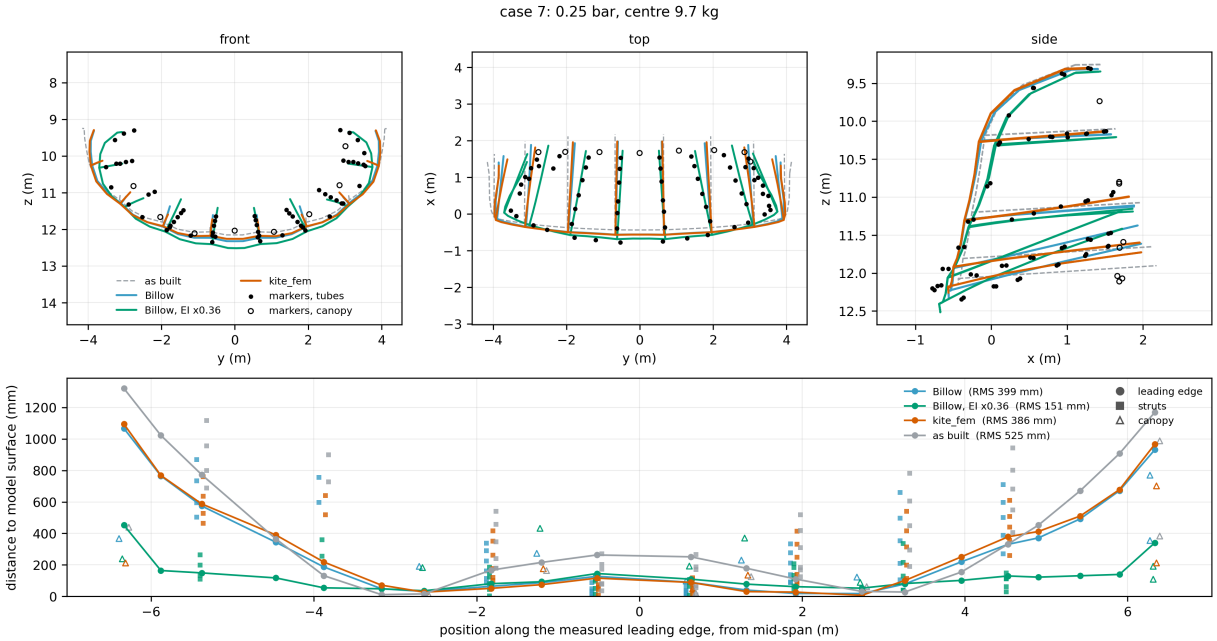


Figure 18: Load case 7.

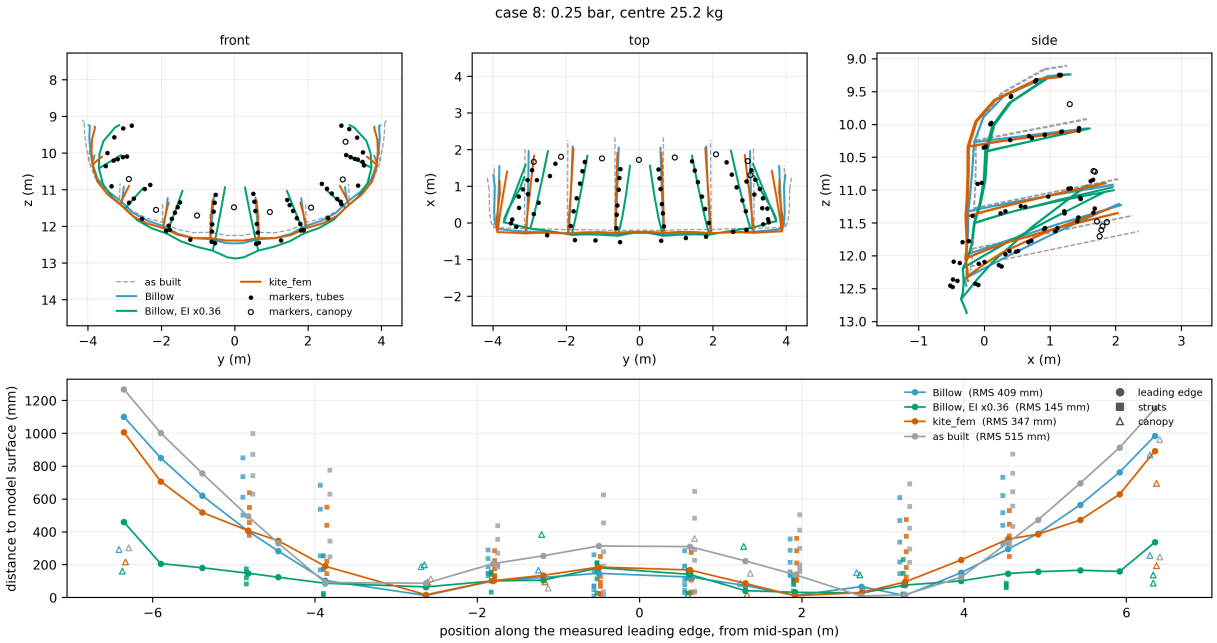


Figure 19: Load case 8.

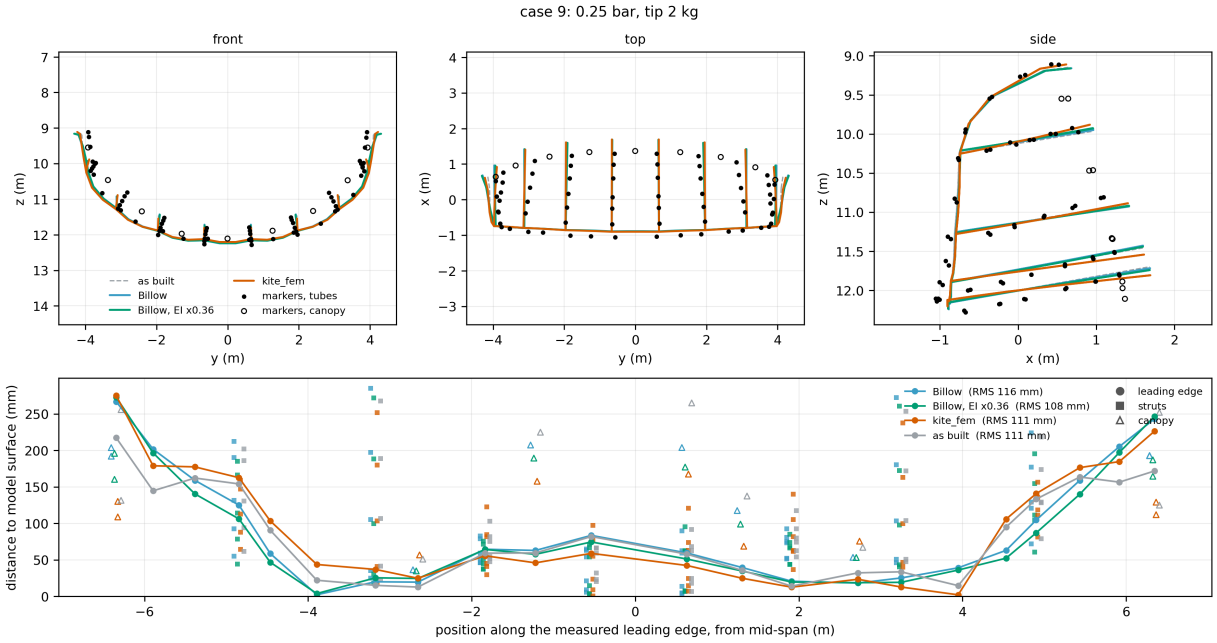


Figure 20: Load case 9.

5.4.3 The centre-load cases, which neither code reproduces

Cases 2, 3, 7 and 8 put 9.7 or 25.2 kg on the mid-span leading edge — a node with *no bridle attached*, carrying up to twice the kite's own weight. Measurement has the span slightly *widening* against bare gravity (5.04 against 4.87 m): the kite tilts forward and the tips fold out.

Neither code reproduces that. `kite_fem` overstates the span by 34–48%, and its author is explicit about the cause: "by applying such a local load directly on the inflatable beam, the cross-section deforms and the stiffness of the tube reduces … the simplification made by modelling such a complex structure as a beam element cannot capture this" (thesis §7.1). Billow fails differently and more visibly: it converges — to residuals of order 1×10^{-8} N — onto a *folded* branch, closing the span to under 1 m. Seeding from the gravity equilibrium and ramping the load in does not recover the physical branch. The energy is non-convex, and this is a second local minimum, not a numerical failure: a converged wrong answer.

These four cases are marked † in Table 7 and reported separately rather than averaged in, because a mean over them measures how each code fails rather than how well it works.

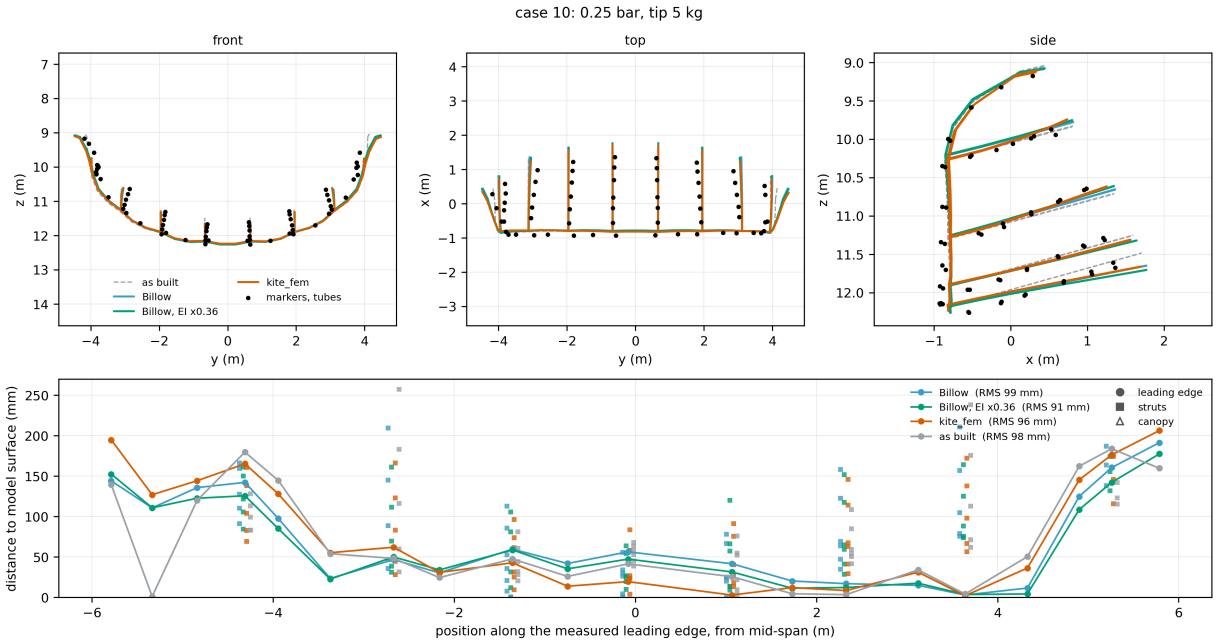


Figure 21: Load case 10.

5.5 What this case actually tests

The comparison was built to isolate the canopy formulation, since that is the one element type that differs between the two codes. It does not do that. A sensitivity sweep on case 1:

change	span (m)
baseline	7.19
canopy Et 5000→1000 N/m	7.15
canopy deleted entirely ($Et = 1$ N/m)	7.36
tube moment–curvature curve scaled by 0.3	**3.73**

Deleting the canopy outright moves the answer less than a quarter as far as a modest change in tube bending. **This is a tube-bending test**, and any canopy conclusion read off it would be unsupported.

Collapse. None of the three codes models post-collapse behaviour. Breukels' source data characterises it — the measured moment falls past collapse — but `kite_fem` computes a collapse flag and never acts on it (a literal `TODO: verify collapse` in `BeamElement.py`), and Billow saturates at $M_{\max}$ instead of dropping. A law that does follow collapse, fixed uniquely by requiring it to match both the fitted initial stiffness and the fitted collapse curvature,

$$W(\kappa) = EI_0\,\kappa_c^2\left[1 - \left(1 + \tfrac{\kappa}{\kappa_c}\right)e^{-\kappa/\kappa_c}\right], \qquad M(\kappa) = EI_0\,\kappa\,e^{1-\kappa/\kappa_c}, \tag{21}$$

converges without difficulty (56–77 iterations, ∼3 s, residual 3×10^{-9}) and changes the answer by 20 mm. It changes nothing here because Billow's beams reach only 31% of the collapse curvature and none of the 98 is flagged collapsed. Note also that collapse arrives at $0.7\,\kappa_0$, i.e. at only ∼52% of $M_{\max}$, so the saturating plateau is already too generous well before the cliff. It matters in general; it does not matter in this dataset.

5.6 Cost, convergence, and what the `kite_fem` numbers are

Table 9: Solver cost. * marks cases that never reached `kite_fem`'s own 0.01 tolerance. **The two time columns were measured on different machines** and are not a controlled benchmark — see the caveats below.

LC	`kite_fem`		Billow		
	t (s)	final tol.	t (s)	iterations	$\|r\|$ (N)
1	546	0.010	7	89	2e-07
2	671	0.010	15	200	9e-08
3	465	0.229*	19	228	4e-07
4	443	0.010	16	212	1e-07
5	619	0.012*	18	240	4e-07
6	576	0.010	6	86	4e-09
7	621	0.010	15	209	4e-09
8	468	0.076*	17	235	6e-09
9	483	0.010	15	187	1e-07
10	626	0.011	16	227	4e-07
mean	552		14	191	

Billow reaches force balance to better than 5×10^{-7} N on every case, against a target of 1×10^{-4} N, in a mean of 14 s. `kite_fem`'s recorded times average 552 s and three of its ten cases stop above its own tolerance (0.229, 0.076, and a marginal 0.0117).

Robustness required one addition to the solver. On slack-dominated configurations a near-zero-curvature search direction is answered with a step of order 1×10^4, the objective overflows, and the line search collapses before restoration can help. A move limit — a trust region expressed as bounds IPOPT already enforces — cures this and reproduces the direct solve to the digit. One caveat came with it: once bounds are active, IPOPT reports success for the *boxed* problem while the iterate sits on the boundary, so the acceptance test must key off the force residual and never off the solver's own verdict.

5.6.1 Caveats on every `kite_fem` number in this section

These matter enough to state plainly, because the comparison is the weakest part of this work and the reader should discount it accordingly.

It was never reproduced. Every `kite_fem` figure here comes from its *committed* `model_results.csv` and saved `.npz` states, not from a run performed for this study. One attempt was made; it produced no output in over ninety minutes and was abandoned. So nothing here distinguishes "`kite_fem` as published" from "`kite_fem` as it runs today".

What *was* verified is narrower but load-bearing: our length extractor reproduces its published table from its own saved states to 1×10^{-9} m on all ten cases (a committed test). The rescoring is therefore sound even though the solve is not reproduced.

Its solver settings are tuning choices we did not replicate. The published run uses an `I_stiffness` of 15, a tolerance of 0.01, a step limit, a relaxation schedule, and an outer loop that calls `adapt_stiffnesses` — progressively stiffening any element above 1% strain, up to four times. None of that was re-derived or assessed here. A different setting could move its answers by an unknown amount.

It is not solving the model it states. Its converged beams sit at about 0.23× its own constitutive law, with negative bending stiffness in three of the four cases inspected, and its anchor node belongs to no element so its equilibrium is held by `I_stiffness` rather than by a load path — its own solutions drift 0.29–0.34 m bodily. Comparing against it is therefore comparing against a numerical state, not a clean reference solution.

"`kite_fem`" is not one thing. The results used here are from `main` at `630c8c0`. AWETrim's own environment installs the branch `mass_proportional_regularisation` at `6edff17` — named for the very regularisation implicated above. Any statement in this section about `kite_fem` refers to the former.

The bridle is not matched in the headline comparison. `kite_fem` ran the bridle exactly as stored. Our primary results apply the Table B.1 corrections. Scored on the stored bridle for a like-for-like comparison, Billow gives 15.1% against `kite_fem`'s 12.4%; on the corrected bridle it gives 15.7%. The difference the bridle makes overall is therefore 0.6 percentage points — small, but the comparison is not bridle-matched and should not be read as though it were.

The metric is ours, not theirs. Its published `shapecorrelation` is not used, for the reason given in Table 1's caption. Numbers here will therefore not match those reported upstream.

The timings are not a benchmark. Its recorded times are from an HP Zbook Power G7; ours are from a different machine, running Python against a compiled CasADi/IPOPT stack while `kite_fem` runs Cython `pyfe3d` elements. Read the ratio as an order of magnitude at most, and note that the two codes are not even solving to the same tolerance — a factor of 1×10^3 separates the residuals.

5.7 What stiffness the measurement implies

Rather than fit bridle lengths, the whole tube moment–curvature curve was scaled by a single factor λ — "`moment_max`", "`bending_stiffness`" and "`torsion_c1`" together, so $\kappa_0 = M_{\max}/EI_0$ is unchanged and the curve keeps its shape while only its magnitude moves. The two bare-gravity cases, the only two that genuinely load the tubes, then say what stiffness the measurement wants:

λ	case 1, 0.15 bar		case 6, 0.25 bar	
	span (m)	error (%)	span (m)	error (%)
1.00	7.276	25.9	7.468	19.5
0.55	6.156	14.6	6.589	11.5
0.45	5.569	9.3	6.120	7.7
0.35	**4.674**	**8.3**	**5.376**	**5.5**
0.25	3.376	18.6	4.141	14.5
measured	4.867		5.453	

Both cases imply $\lambda \approx 0.36$, and — this is the part that matters — they imply the *same* factor at both pressures. A pressure-dependent error in the fit would not do that; a constant offset would. It is also further evidence against the low-pressure extrapolation of $M_{\max}$ discussed above, which would have required more softening at 0.15 bar than at 0.25 bar.

Run across all ten cases, a single $\lambda = 0.36$ gives a mean relative error of 8.0% against `kite_fem`'s 12.4%, and 6.2% against 11.5% on the two cases that test the tubes. It also repairs the centre-load cases, which the unscaled model got worst: case 2 falls from 27.6% to 9.6%, case 7 from 19.7% to 7.5%. The tip-load cases barely move (6.5%→6.6%, 5.2%→5.2%), exactly as the previous section predicts — they are bridle-determined and carry no information about the tube law.

This is an identification, not a fit. One scalar, applied uniformly to all 98 tube elements, calibrated on two cases and improving eight. It is reported because a consistent, pressure-independent factor is evidence about the fit rather than a free parameter absorbing error — but it is not adopted as a model change. Two things must be understood first.

Where does 0.36 *come from?* Inverting the fitted EI_0 through $EI = Et\pi r^3$ gives an implied Et of 133–155 kN/m for the leading-edge tubes, nearly constant across $d = 0.113 - -0.202$ m — so the fit's radius dependence is physically consistent, and the implied fabric (0.5–1.2 GPa at 0.15–0.25 mm) is plausible to soft for polyester. The fit is therefore not obviously twice too stiff on material grounds, which makes a calibration or usage mismatch the more likely explanation.

Two candidates are worth checking against Breukels' thesis directly. First, his tubes are discretised as *rigid cylindrical segments joined by torque springs*, and the C_1–C_8 coefficients were fitted so that such a chain reproduces the measured beam — a discretisation-dependent calibration, not an intrinsic moment–curvature law, and re-using it as a continuum law could carry a systematic factor. Second, his nomenclature defines an *effective* internal pressure p_{eff} distinct from the gauge pressure; if the fits are calibrated on p_{eff}, feeding 0.15 bar gauge straight in is wrong at exactly this kind of magnitude.

5.8 How weakly the published data constrains a model

The twelve published quantities are less information than they look. Every left/right pair is identical to 0.0×10^0 — `La`≡`Li`, `LcsTL`≡`LcsTR` — so the data is exactly mirrored and only six numbers are independent. The thesis itself scores just three: the trailing-edge length (the sum of `La`–`Li`), the span, and the tip-to-centre distance.

More limiting is what is absent. There is no height, no sag depth, and no chordwise measure of any kind: every published quantity lies in the spanwise / trailing-edge geometry. A model can

match all three scored lengths and still have the wrong shape — the wrong camber, the wrong chordwise twist, the wrong depth of sag — with nothing in the dataset able to detect it. That is a real ceiling on what any validation against this file can claim, including this one.

The information exists. The measurement was stereoscopic, giving full three-dimensional marker positions, and the thesis plots measured geometry against model geometry in top, side and front projection for all ten load cases (its Figure 7.2). Only the twelve scalars were published with the model.

The 3D marker coordinates behind that figure turn this from a comparison of three effective scalars into a comparison of shapes. They are published in Haanen's repository, vendored here, and used in Section 5.9; the length comparison above is kept so that the published numbers can be reproduced, not because it can decide a shape.

5.9 The proper test: 3D shape against the markers

Everything above is scored on twelve numbers that reduce to six independent ones. Haanen's marker clouds — 62 to 76 three-dimensional points per load case on the leading edge, the eight struts and the canopy, vendored under `validation/data/hanging_validation/markers/` (MIT) — allow the question that data cannot answer: is the *shape* right? Every model is scored the same way: Billow, `kite_fem`'s published states, and the undeformed as-built geometry as the null model.

Method. Markers are in the stereo rig's frame, so a *rigid* transform is needed — rotation and translation only, never scale, since scale is what is being tested. Correspondence comes from structure rather than fitting: the measured struts are ordered across the span with their first marker at the leading edge, so their roots pair with the model's eight strut leading-edge nodes, giving eight anchors for a Kabsch fit. Both spanwise orderings are tried, because the kite is near-symmetric and the marker numbering direction is undocumented. Each model gets its own fit; none is scored in a frame chosen for another. The fit is refined by ICP against the model tube axes, so no marker is assumed to sit on a node, and finished by Gauss–Newton, with a backtracking line search, on the quantity that is reported: the distance from each marker to the tube *surface*. The markers are stuck to the outside of the tubes while the model polyline is the tube axis, and ignoring that radius — 55 to 100 mm here — would put a systematic offset into every case. Both stages are needed. Fitting to the axis alone biases every model toward the markers by up to a radius, because the markers all sit on one side; fitting to the surface from a rough start can lock onto the far side of a tube, a diameter away, where the distance is zero again. A committed test places synthetic markers on the model's own surface, moves them by an arbitrary rigid motion, and requires the fit to recover them to under a millimetre.

The canopy markers take no part in the fit. They are scored afterwards against the triangulated canopy surface, as an independent check on the alignment and on the fabric.

The scaled model reaches the measurement floor. On the two cases that load the tubes, the error falls from 480 to 100 mm and from 387 to 100 mm — onto the same ∼100 mm the tip cases sit at, which is the marker placement and alignment floor. Across all ten the mean falls from 314 to 124 mm, against 403 mm for doing nothing and 242 mm for `kite_fem`. That the factor identified from two scalar spans also fixes the full three-dimensional shape, including the centre-load cases it was not fitted on, is the strongest evidence in this section that it is a real property of the tube law rather than a number absorbing error.

`kite_fem` on the same footing. Scored by the identical procedure, `kite_fem` sits between the two Billow runs on the gravity cases (171 and 159 mm), consistent with its beams converging to a fraction of their own law. On the centre-load cases it is barely better than the null model

Table 10: Point-to-surface RMS (mm) between each solved shape and the photogrammetry tube markers, each model under its own rigid fit. "As built" is the undeformed geometry, included as the null model: anything not beating it has explained nothing. † centre-load cases. Generated by `run_hanging_shape_comparison.py`.

LC	load	Billow	Billow, EI x0.36	kite_fem	as built	markers
1	gravity	480	100	171	695	65
2†	centre 9.7 kg	501	152	465	654	65
3†	centre 25.2 kg	506	171	452	639	62
4	tip 2 kg	142	134	138	139	64
5	tip 5 kg	98	89	93	99	64
6	gravity	387	100	159	554	65
7†	centre 9.7 kg	399	151	386	525	61
8†	centre 25.2 kg	409	145	347	515	63
9	tip 2 kg	116	108	111	111	66
10	tip 5 kg	99	91	96	98	62
mean, gravity only		434	100	165	625	
mean, centre load		454	154	412	583	
mean, tip load		114	105	110	112	
mean, all ten		314	124	242	403	

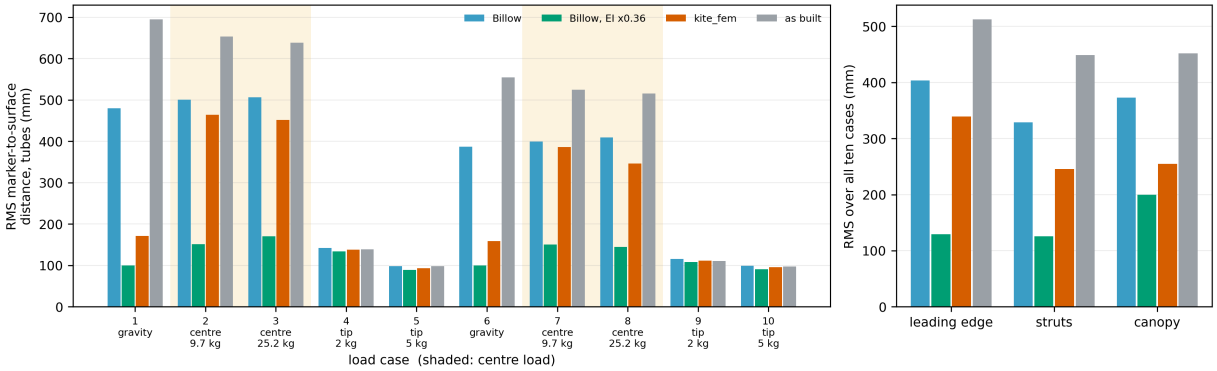


Figure 22: Left: Table 10 per load case. Right: the same distance pooled over all ten cases and split by marker group; the canopy markers are not used in the fit, so their column is an independent check.

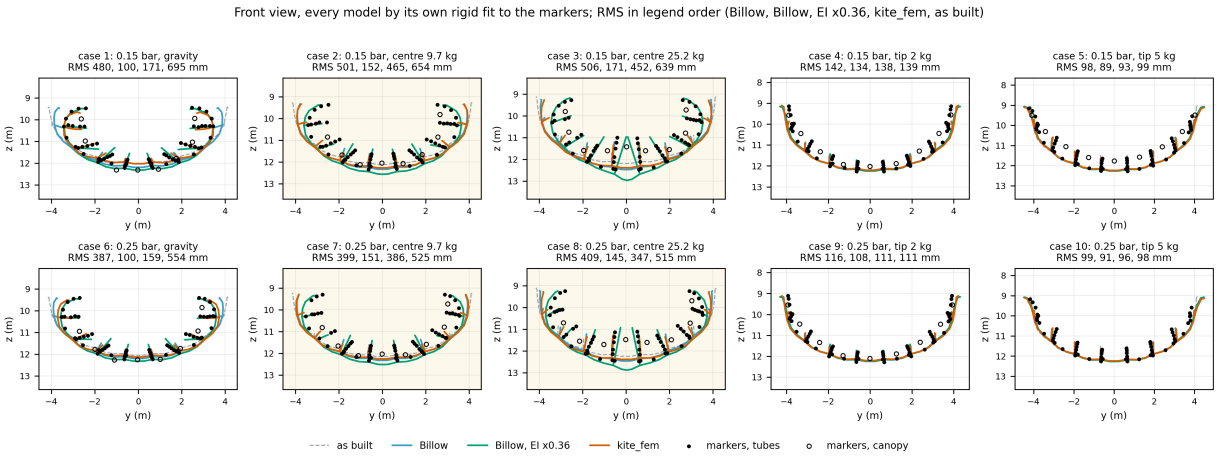


Figure 23: Front view of all ten cases, every model under its own rigid fit to the markers. Shaded panels are the centre-load cases.

(347–465 mm against 515–654 mm), which is its author's own diagnosis seen in three dimensions: a beam element cannot follow the local cross-section collapse under a concentrated load.

The tip-load cases are confirmed uninformative. Their errors are 139, 99, 111, 98 mm for the *undeformed* geometry against 142, 98, 116, 99 mm for the unscaled model, 134, 89, 108, 91 mm for the scaled one and 138, 93, 111, 96 mm for `kite_fem` — all within about 10 mm of each other. Doing nothing scores as well as either model. This is the same conclusion the bridle-tension argument reached in Section 5.5, now established directly in three dimensions rather than inferred, and it is why the six-case and ten-case means elsewhere in this section should be read with care.

5.10 Summary

- The distributed reference model is ill posed — its anchor belongs to no element — and its published results are held up by a numerical regularisation. Any future use of this dataset has to supply a suspension.
- `kite_fem`'s beams converge to $\sim 0.23\times$ its own constitutive law, with negative bending stiffness in several cases. Its closer agreement with the measurement is not evidence of better physics.
- The dataset is a tube-bending test — and only two of its ten cases even are. It cannot decide a canopy model, which was the original reason for building the comparison. The 3D markers confirm this directly: on the tip-load cases the undeformed geometry scores as well as either model.
- Against the 3D markers, $\lambda = 0.36$ takes the mean point-to-surface error from 314 to 124 mm (null model 403 mm, `kite_fem` 242 mm), and brings the two tube-loaded cases onto the $\sim$100 mm measurement floor — having been identified on two scalar spans, not on the shape.
- Post-collapse behaviour is unmodelled in all three codes and does not matter here or in coupled use, because the tubes never leave their initial slope.
- The residual disagreement lives in EI_0. A single pressure-independent factor $\lambda \approx 0.36$, identified on the two gravity cases, improves eight of the ten and beats `kite_fem` overall (8.0% against 12.4%). It is reported as an identification, not adopted: two specific mismatches in how the Breukels fit is being re-used — its torque-spring discretisation and its effective pressure — should be checked against the source first.

6 Demonstration cases

6.1 Wrinkling: fabric in a frame smaller than itself

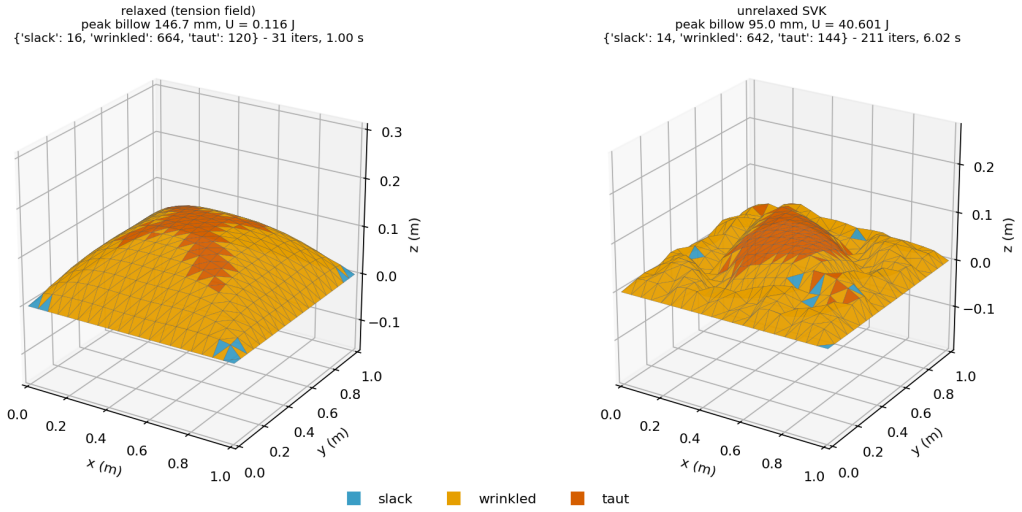


Figure 24: A panel clamped into a frame 6% smaller than itself, then pushed out of plane. The relaxed law converges in 31 iterations to a smooth billow; the unrelaxed one takes 211 and buckles into mesh-scale crumple at less than two-thirds the deflection.

This is the clearest illustration of why the relaxed energy is not optional. The unrelaxed panel stores 40.6 J against the relaxed panel's 0.116 J, almost all of it fighting compression that fabric cannot carry.

6.2 Large-deflection cantilever

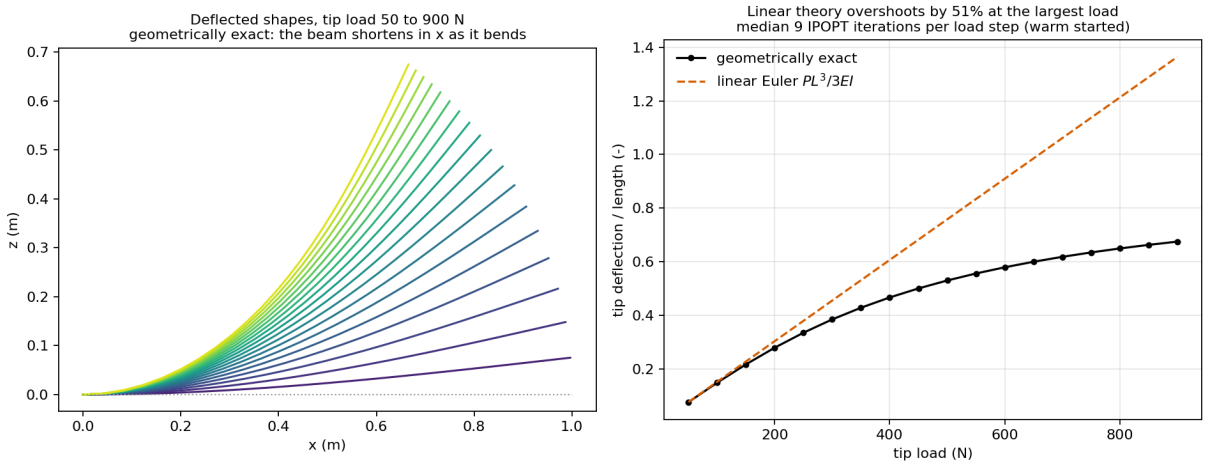


Figure 25: Deflected shapes under a tip load swept from 50 to 900 N, and tip deflection against linear theory. The beam visibly shortens in x as it bends — the geometric nonlinearity linear theory omits. At the largest load, linear theory overshoots by 51%.

Median 9 IPOPT iterations per load step with warm starting.

6.3 Batten-stiffened sail

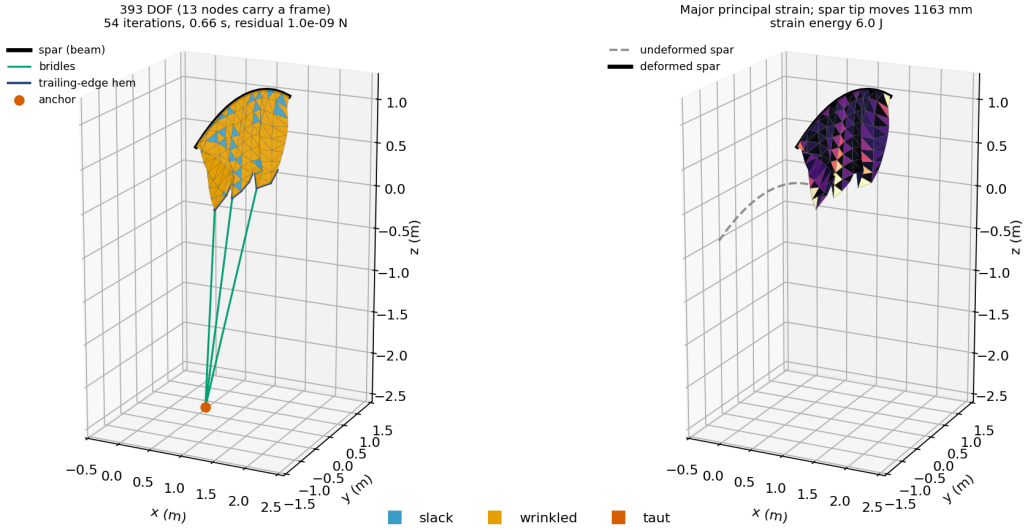


Figure 26: Beams, membranes, cables and a trailing-edge hem in one energy: a spar carrying bending and torsion, a wrinkling canopy hanging off it, three bridles to a fixed anchor. 393 DOF, 54 iterations, 1.4 s, residual 1.0×10^{-9} N.

Note that only 13 of the nodes carry a material frame; the canopy nodes stay at three degrees of freedom.

6.4 Scaling

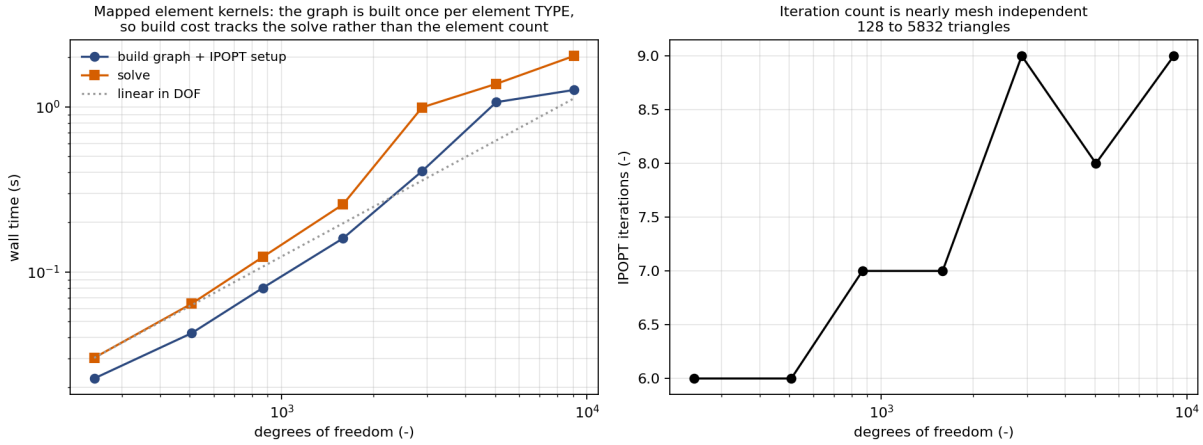


Figure 27: Build and solve time against problem size, and iteration count. Because the graph is built once per element *type*, build cost tracks the solve rather than the element count, and the iteration count is nearly mesh independent across a 37× range in degrees of freedom.

6.5 Ram-air canopy

Six box cells side by side, top and bottom skins, leading- and trailing-edge closure walls and a rib on every cell boundary, on bridle lines to one tow point (`examples/run_ramair_demo.py`, Fig. 28). The internal pressure is one uniform gauge pressure over the outer envelope, applied as the potential $-pV$; the internal ribs see the same pressure on both faces, carry no net pressure load, and are plain membranes in no pressure set. The skins are cut 2% oversize in span so they pillow between the ribs. Pressure alone has zero resultant, so a dead lift on the top skin stands

in for the aerodynamic load and puts the bridles to work. Pressure and lift are ramped together from 60 to 300 Pa on one solver build, each step warm-started from the last: 141, 18, 26, 14 and 6 iterations, every step converged with the acceptance tolerance scaled to the nodal pressure force. The box contracts in span by 17% because the pillowing skins pull the outer ribs inward harder than the pressure on them pushes out, and the ribs read wrinkled throughout, which is what a membrane in uniaxial vertical tension is.

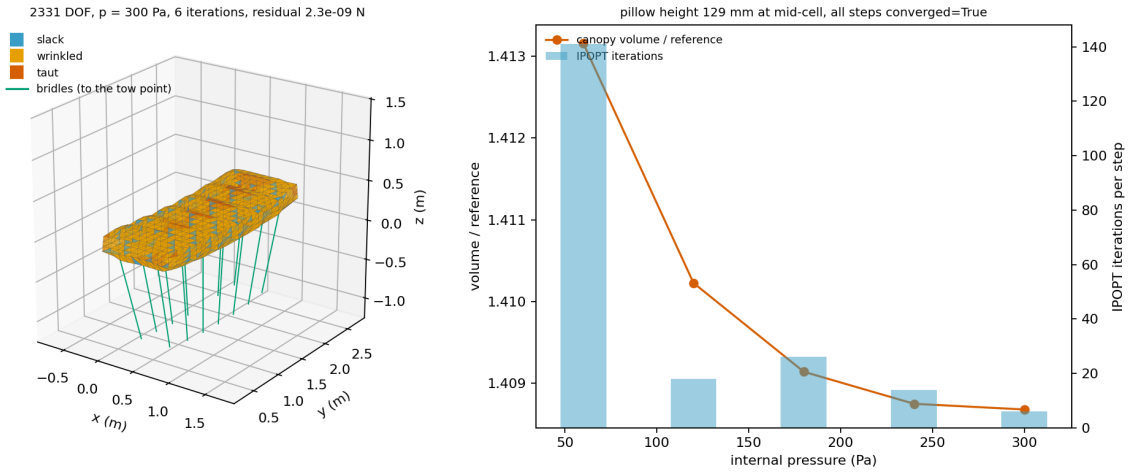


Figure 28: Ram-air canopy under one uniform pressure. Left: the solved canopy at 300 Pa coloured by membrane regime, bridles running to the tow point below. Right: canopy volume against reference and IPOPT iterations per ramp step.

7 Comparison with `kite_fem`

`kite_fem` (with `pyfe3d`) is the existing FEM path in the AWETrim tree. This section states what differs, what was measured, and where the comparison is and is not fair.

7.1 Capability

	`kite_fem` / `pyfe3d`	**Billow**
Elements	springs, pulleys, inflatable Timoshenko beams	cables, pulleys, geometrically exact Timoshenko beams, inflatable tubes, wrinkling membranes
Canopy fabric	noncompressive spring net (no membrane element exists)	CST membrane, Pipkin relaxed energy
Degrees of freedom	6 per node everywhere, masked via `bc`	3 per node, +3 only on beam nodes
Formulation	Newton–Raphson on $\mathbf{f}_e - \mathbf{f}_i$	minimum potential energy (IPOPT)
Assembly	scipy sparse COO per element	one mapped kernel per element type
Regularisation	`I_stiffness=25`, `step_limit=0.2`, `relax=0.5`, optional pseudo-transient	IPOPT inertia correction; membrane slack blend
Warm start	none (`solve` resets to undeformed)	state and duals
Default tolerance	10^{-2} N	10^{-8} IPOPT, accepted at 10^{-6} N
Inflatable tube	secant EI/GJ plus collapse flag	the same fits integrated to $W(\kappa)$
Experimental validation	hanging-kite test against measured data	none

7.2 Two traps in making it a fair comparison

1. `FEM_structure.solve` defaults to `I_stiffness=25`, an **absolute** regularisation in N/m added to every degree of freedom of the tangent. The benchmark beams here have $EI/L^3 = 10\,\mathrm{N/m}$, so the default is larger than the structure: 2000 iterations with the residual stuck at 20 N. With it at zero the same problem converges in 38. It is a kite-scale tuning constant, not a universal one. All results above use zero.
2. The property matching was verified in the linear limit before any conclusion was drawn: below about one degree of rotation the two codes agree to 0.02%.

7.3 Cost and accuracy on a common problem

Tip-loaded slender cantilever, matched properties, same mesh, one machine, one run.

Table 11: Source: `run_cost_comparison.py`. "Warm" is a second solve at a 5% different load; `kite_fem` restarts from the undeformed configuration on every call, so its warm column is a second cold solve.

α	El.	Model	DOF	Setup (s)	Cold (s)	Warm (s)	Iter.	ms/iter
1	20	Billow	126	0.133	0.249	**0.081**	16	15.6
1	20	`kite_fem`	126	**0.004**	**0.132**	0.151	39	**3.4**
1	80	Billow	486	0.217	0.733	**0.282**	17	43
1	80	`kite_fem`	486	**0.009**	**0.189**	0.192	39	**4.8**
6	40	Billow	246	0.076	**0.333**	**0.062**	**22**	15
6	40	`kite_fem`	246	**0.005**	1.384	1.526	444	**3.1**
6	80	Billow	486	0.186	**0.669**	**0.233**	**23**	29
6	80	`kite_fem`	486	**0.005**	2.270	2.658	490	**4.8**

`kite_fem` has roughly 6× cheaper iterations (compiled `pyfe3d` plus scipy sparse) and 30× faster setup. Billow uses 2–20× fewer iterations, and its count is nearly independent of both mesh and load (16–23 throughout) where `kite_fem`'s climbs from 39 to about 490 as the load

grows. `kite_fem` therefore wins on easy, small, one-shot problems; Billow wins as soon as the problem is hard, reused, or large. Billow's setup cost is a one-off amortised across a sweep — exactly the coupled-loop pattern.

7.4 The canopy model: spring net versus membrane

Since `kite_fem` has no membrane element, its canopy is a noncompressive spring net (chordwise and spanwise links, *no diagonals*; its own example carries a commented-out block of "fictional springs constraining the canopy from collapse"). Billow can build exactly that net from its cable elements, which makes a genuine like-for-like study possible.

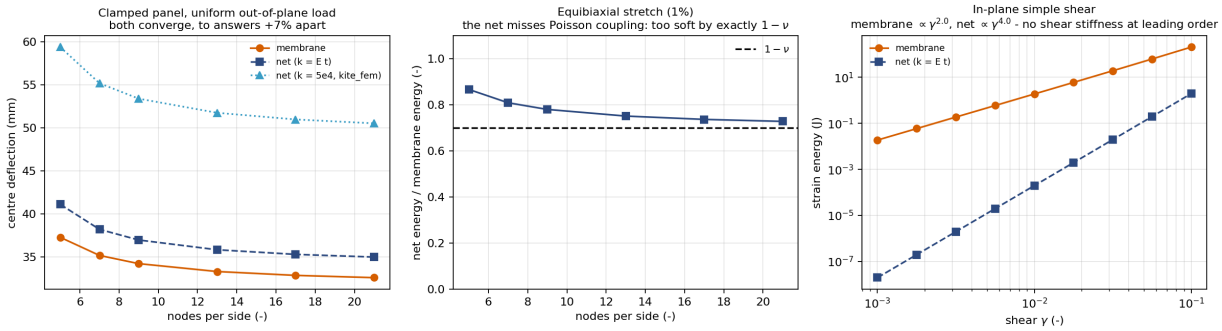


Figure 29: Spring net versus CST membrane on identical meshes, both minimised. Left: both converge under out-of-plane load, to answers 7% apart. Middle: under equibiaxial stretch the net is too soft by exactly $1-\nu$. Right: under in-plane shear the membrane energy scales as $\gamma^{2.0}$ and the net as $\gamma^{4.0}$ — the net has *no shear stiffness at leading order*.

Measured: biaxial energy ratio 0.728 against $1-\nu = 0.70$; shear exponents 2.018 (membrane) and 3.999 (net).

7.4.1 Taut: shape and cost

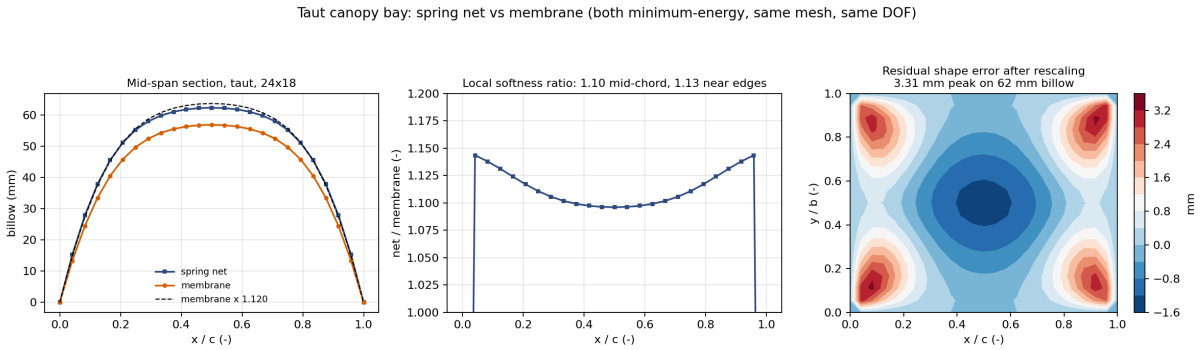


Figure 30: Taut bay, 24×18, both minimised on the same mesh with the same 1425 degrees of freedom, so this is purely the canopy model. The net billows 9.6% deeper; the best single scale factor is 1.120, and a 3.31 mm residual shape difference remains after removing it.

The local softness ratio is not constant: 1.10 at mid-chord rising to 1.145 near the clamped edges, where the fabric carries biaxial tension and shear that the net cannot represent. The residual map shows the net relatively too deep in the corners and too shallow in the middle — it billows into a slightly different surface, not merely a deeper one.

The scale factor cannot be calibrated away, because it depends on the stress state: about 1.10–1.12 when taut, but 1.00 when fully wrinkled (the slack cases agree within 1%, since a

deeply slack panel carries essentially uniaxial chordwise tension where the two models coincide). The correction needed would change with flight condition.

Cost is not a differentiator. At 24×18: net 0.092 s setup and 0.227 s solve in 12 iterations; membrane 0.176 s and 0.245 s in 8. The membrane's per-iteration cost is higher but it needs fewer iterations, and the two roughly cancel.

7.4.2 Robustness

Table 12: Peak billow (% chord) and section RMS against each model's own finest mesh. The net fails outright at one intermediate slack mesh — the signature of its zero-energy shear mode. The membrane never failed.

	Spring net			Membrane		
Mesh	Camber	RMS	Conv.	Camber	RMS	Conv.
taut						
4×3	2.858	0.388	yes	2.542	0.401	yes
8×6	3.129	0.051	yes	2.842	0.051	yes
16×12	3.117	0.010	yes	2.842	0.010	yes
24×18	3.115	—	yes	2.842	—	yes
slack 3%						
8×6	7.388	0.186	yes	7.449	0.143	yes
12×9	7.240	0.091	yes	7.294	0.097	yes
16×12	5.675	1.719	**no**	7.412	0.031	yes
24×18	7.302	—	yes	7.402	—	yes

7.4.3 Canopy model versus solver, separated

A single bay isolates the two effects that otherwise get conflated. Rows 2 and 3 solve the *identical* spring net, so their difference is the solver alone.

Table 13: One 2.0×1.3 m bay at 250 Pa, all edges fixed.

Case	Configuration	Solve (s)	Iterations	Camber (% c)	Converged
taut 8×6	Billow + membrane	0.030	7	2.842	yes
taut 8×6	Billow + net	0.026	10	3.129	yes
taut 8×6	`kite_fem` + net	10.04	2729	3.129	yes
taut 12×9	Billow + net	0.034	9	3.083	yes
taut 12×9	`kite_fem` + net	22.00	3001	5.283	**no**
slack 8×6	Billow + net	0.049	23	7.388	yes
slack 8×6	`kite_fem` + net	9.06	3001	7.363	**no**

Where both converge they agree exactly (3.129% both), which is what makes the `kite_fem` driver used throughout this document trustworthy. On the identical net, minimum energy is roughly 390× faster than the damped Newton solve, and remains convergent where the Newton solve does not.

7.5 Full kite

A V3-scale structure — inflated leading-edge tube, inflated struts at every bay boundary, canopy over nine bays, trailing-edge hem, bridle to a fixed anchor, uniform 250 Pa — was built with the same nodes, tubes and cables in both codes.

Model	DOF	Setup (s)	Cold (s)	Warm (s)	Iter.	Residual (N)
Billow	1893	0.61	**5.91**	**0.58**	73	7.2×10^{-9} (converged)
`kite_fem` (I=25)	2976	0.05	76.2	72.0	2001 (cap)	4.1×10^{5} (diverged)
`kite_fem` (I=0)	2976	0.06	108.9	68.9	2001 (cap)	1.3×10^{8} (diverged)

Interpretation, carefully. No usable `kite_fem` full-kite number was obtained. Diagonals, canopy curvature, both regularisation settings and load stepping with rebuilt initial conditions were all tried; all diverged, with residuals roughly 100× the applied load. **This is consistent with a known open issue on the `kite_fem` side** rather than a defect in the driver used here — the ASKITE integration commit records "kite_fem still diverging, but not due to the coupling". The only kite-scale figure that comes from its own workflow is its hanging-kite validation, at 442–671 s per load case. The single-bay comparison above is the trustworthy cost comparison.

8 How many membrane triangles?

Chordwise and spanwise resolution are swept independently, because they are not the same problem: a canopy bay billows across the chord — which sets the section the aerodynamics sees — while spanwise it varies slowly between struts.

The test case is one bay of the LEI-V3 canopy: chord 2.0 m, strut spacing 1.3 m (the V3 spans 8.28 m over 10 stations), held on all four edges, under 250 Pa. Both a taut panel and a realistically slack one (reference cut 3% oversize) are run.

Convergence is judged on quantities the coupling actually consumes: peak billow over chord, and the RMS deviation of the mid-span chordwise profile from the finest mesh. **Peak local strain is deliberately not a criterion**: in a wrinkling membrane it does not converge, and nothing downstream reads it.

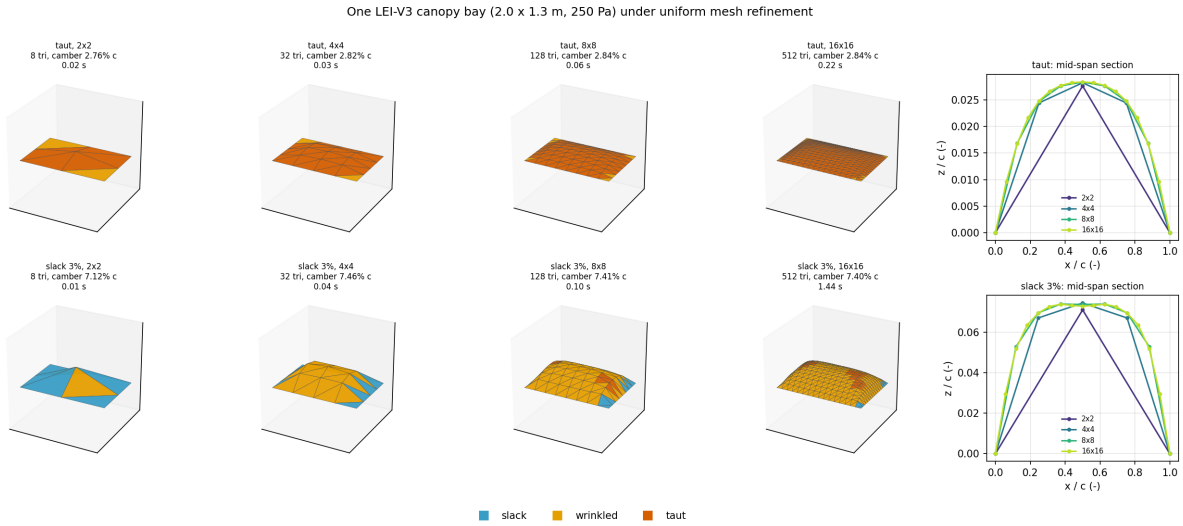


Figure 31: One bay under uniform refinement. A 2×2 bay is a tent, not a canopy: the camber is only 2.8% low but the section is two straight lines. By 8×8 the profile is visually converged.

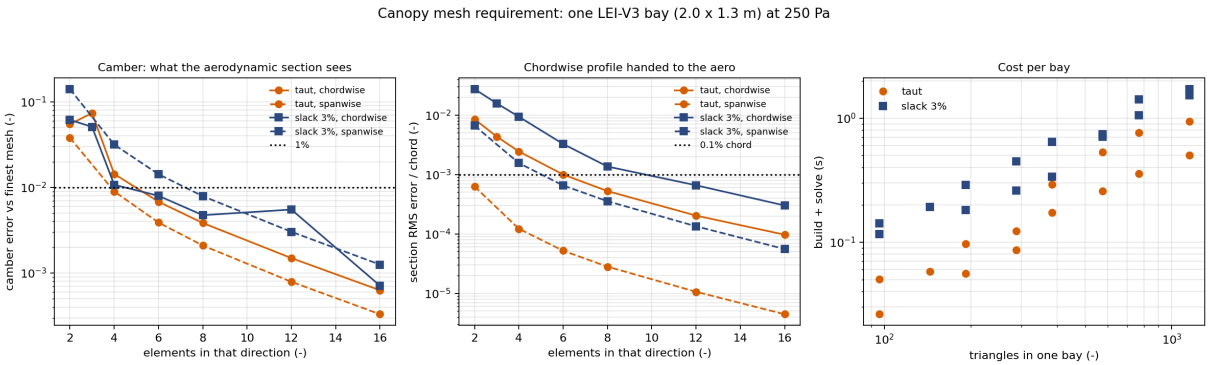


Figure 32: Camber and section-shape convergence, chordwise (solid) and spanwise (dashed), and cost per bay.

Table 14: Candidate meshes against a 24×24 reference, slack 3%.

Mesh	Triangles	Camber error	Section RMS (% c)	Build + solve (s)	Iter.
4×4	32	0.88%	0.967	0.03	12
6×6	72	0.61%	0.338	0.07	17
8×6	**96**	**0.71%**	**0.145**	**0.09**	19
8×8	128	0.16%	0.140	0.17	15
12×8	192	0.18%	0.075	0.18	21
16×12	384	0.21%	0.033	0.44	22

Recommendation. **8 chordwise × 6 spanwise per bay**, 96 triangles, about 900 for the whole nine-bay canopy. Camber error 0.71% and section RMS 0.145% of chord — 2.9 mm on a 2 m chord, comfortably below what an airfoil parameterisation resolves. Use 12×8 for final runs: it halves the shape error for twice the cost.

Slack costs roughly twice the mesh. At 3% oversize the panel is 88% wrinkled (against 1–4% taut) and needs about 12×8 for the same accuracy, with 4× the iterations. Since a real canopy flies slack, the as-cut slack of the actual sail drives the mesh requirement more than anything else here and is worth pinning down.

Caveat. These numbers are for *uniform* pressure. A real chordwise pressure distribution is peaked near the leading edge and may demand more chordwise resolution; the study should be repeated once loads come from an aerodynamic solve.

9 Known limitations

The measured validation cannot decide the canopy. Section 5 scores Billow against a real kite, but that dataset is governed by inflatable-tube bending: deleting the canopy entirely moves the span by 0.17 m while a modest change in tube stiffness moves it by 2.5 m. So the claim that a wrinkling membrane is more correct than a spring net still rests on it being the physically appropriate model, not on measurement. A test that loads the canopy in its own right — pressure normal to a restrained bay, with the deformed surface measured — is the missing experiment.

Tube stiffness is unresolved at roughly a factor of three. The same section identifies $\lambda \approx 0.36$ on the moment–curvature curve, which reproduces the measured three-dimensional shape but is *not* adopted: the likelier cause is a mismatch in how the Breukels fit is re-used

rather than a wrong fit. Until that is settled, absolute deflections from any Billow model using `InflatableTubeLaw` — including coupled aero-structural runs, whose tubes sit on the same initial slope — carry that uncertainty.

Applied moments. Need load stepping, and are exact only about a fixed axis; a dead three-dimensional moment is non-conservative and has no potential. Nodal forces are unaffected.

Post-collapse tubes. Reported, not modelled. The fits cover bending and torsion only; axial and shear stiffness must be supplied from tube geometry.

Dead loads. Loads are frozen per solve, which matches a staggered coupling loop. A uniform pressure on a closed or pinned-rim surface is conservative and is a potential ($-pV$); a flow-dependent, non-uniform pressure distribution is not and could not be posed as one, and goes through `lumped_pressure_forces` as a lagged dead load — worth remembering before anyone tries to fold aerodynamics into the same minimisation.

Pressure is prescribed. A fixed-mass gas law, $-p_0 V_0 \ln(V/V_0)$, is a function of the global volume rather than of one element and is not provided; an open ram inlet is a free boundary and not conservative, so inlets are modelled closed. A per-triangle pressure column is refused by the diagnostics but not checked inside the solve.

No contact. Contact and self-contact are not modelled.

Setup cost. About 0.2 s of graph construction for a 486-DOF beam is disproportionate against `kite_fem`'s 0.005 s, even though it is a one-off and stops mattering at membrane scale.

Threaded assembly. `PotentialEnergy(parallelization=...)` exposes CasADi threaded maps but is untested at scale; serial is the default.

Not coupled. There is no aerodynamic coupling adapter yet.

10 Reproducing everything in this document

Figure / table	Command
Figs. 24–27, 29	`python scripts/structural/run_demo_cases.py`
Figs. 1–4	`python scripts/structural/run_validation_benchmarks.py`
Cantilever cost table	`python scripts/structural/run_cost_comparison.py`
Figs. 31, 32	`python scripts/structural/run_mesh_requirement.py`
Single-bay comparison	`python scripts/structural/run_panel_cost.py`
Full-kite table	`python scripts/structural/run_full_kite_cost.py`
Hanging-kite results, Tables	`python scripts/structural/run_hanging_validation.py`
Stiffness scan	`python scripts/structural/run_hanging_stiffness_scan.py`
3D shape comparison	`python validation/run_hanging_shape_comparison.py`
Figs. 5–7	`python validation/run_pressure_benchmarks.py`
Fig. 28	`python examples/run_ramair_demo.py`
Hanging figures	`python scripts/structural/plot_hanging_validation.py`
Generated `.tex` tables	`python scripts/structural/make_hanging_report.py`
Test suite	`pytest tests/structural/`

10.1 Package layout

`rotations.py`	$SO(3)$ kernels over an `xp` namespace (NumPy or CasADi)
`model.py`	`DofLayout`, `StructuralState`, `StructuralModel`
`energy.py`	`PotentialEnergy`: mapped assembly, gradient, tangent stiffness
`solver.py`	`MinimumEnergySolver`, `StructuralSolution`
`elements/base.py`	`ElementKernel` protocol, `ElementSet`
`elements/cable.py`	`CableKernel`, `PulleyKernel`
`elements/beam.py`	`TimoshenkoBeamKernel`, `BeamSection`
`elements/inflatable.py`	`InflatableTubeLaw`, `InflatableBeamKernel`
`elements/membrane.py`	`MembraneKernel`, `membrane_regimes`
`elements/pressure.py`	`PressureKernel`, `pressure_forces`, `lumped_pressure_forces`

10.2 Minimal example

```
import numpy as np
from awetrim.structural import MinimumEnergySolver, StructuralModel
from awetrim.structural.elements import build_cable_elements

nodes = np.array([[0., 0, 0], [1., 0, 0]])
cables = build_cable_elements([[0, 1]], rest_lengths=[1.0], stiffness=[1e4])
model = StructuralModel(nodes, [cables], fixed_translation_nodes=[0])
solution = MinimumEnergySolver(model).solve(np.array([[0., 0, 0], [10., 0, 0]]))
```

10.3 Invariants to preserve

- **Isolation.** NumPy and CasADi only.
- **One mapped kernel per element type.** Never reintroduce a Python loop over elements that builds `MX`.
- **Single source for every formula.** `rotations.py` holds the $SO(3)$ maps against an `xp` namespace so NumPy and CasADi share one implementation; `beam_strains` and `green_strain` are called by both the kernels and the NumPy setup and diagnostic paths.
- **No CasADi across the module boundary.** Symbolics are created inside `energy.py` and stay there.
- **Kernels are pure.**
- `converged` **is the physical verdict**, with IPOPT's own kept alongside.

References

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[8] O. Cayon, M. Gaunaa and R. Schmehl, "Fast aero-structural model of a leading-edge inflatable kite", *Energies* **16** (2023) 3061.

[9] ASKITE, `https://github.com/awegroup/ASKITE`; `kite_fem`, `https://github.com/awegroup/kite_fem`.